

Accessible Technology for Adults with Down Syndrome: A Scoping Review

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This article presents a scoping review of assistive technologies designed to support Adults with Down Syndrome (AwDS), focusing on individuals aged 18 and older. We define assistive technologies as digital tools used directly by AwDS to assist them in performing a task or achieving a goal. The review aimed to: (1) examine how researchers have involved AwDS in the design and evaluation of technologies; (2) identify the types of technologies explored and the motivations driving their use; (3) assess how these technologies support abilities across life domains; and (4) analyze the reported benefits, challenges, and design implications. Following PRISMA-ScR guidelines, we searched six academic databases (ACM Digital Library, IEEE Xplore, Compendex, Scopus, ScienceDirect, and SpringerLink), resulting in 1,944 articles. After applying the inclusion criteria, we identified 20 studies for in-depth analysis. Methods ranged from experimental to participatory design, with inclusive strategies such as caregiver involvement, visual aids, and personalized supports frequently reported. Technologies identified included computers and web platforms, mobile applications, AAC systems, smart home tools, and social robots, supporting areas like daily living, communication, learning, navigation, and health. Reported benefits included increased independence and digital inclusion, enhanced communication, expanded social connection, greater confidence and engagement, and enhanced skill development. Common challenges included interface complexity, lack of context-aware interaction, environmental and sensory constraints, and uneven impact across users. Based on these findings and paper analysis, we outline five design recommendations: reduce interaction complexity, support engagement through feedback and flexible pacing, enable personalization and long-term adaptability, provide accessible pathways to secure system access, and embed technology within everyday routines. This review emphasizes the importance of inclusive, user-centered systems that align with the goals, contexts, and lived experiences of AwDS.

CCS Concepts: • **Human-centered computing** → **Accessibility design and evaluation methods**; **Accessibility technologies**; **Accessibility systems and tools**;

Additional Key Words and Phrases: Assistive technologies, Cognitive disability, Down syndrome, Literature review, Scoping review, Technology outcomes for persons with disabilities

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1 Introduction

Technology is playing an increasingly central role in everyday life, influencing how we communicate, learn, work, navigate spaces, and express ourselves. For many, digital tools can enhance autonomy, productivity, and social connection. Despite its widespread adoption, individuals and groups within our society face barriers to accessing the benefits of technology, highlighting the need to design more inclusive technologies for a wider range of people while including people with disabilities [114, 151]. People with cognitive disabilities, such as **Adults with Down Syndrome (AwDS)**, may face barriers in accessing mainstream, emerging, or assistive technologies. These barriers can stem from cost [78, 141], lack of awareness [53, 78, 121], limited training opportunities [53, 85, 128], or a mismatch between the technology's design and the user's preferences or situated abilities [112, 141, 155]. Whether navigating common technologies, such as smartphones, tablets, and social media, or using specialized systems, including **Augmentative and Alternative Communication (AAC)** devices, smart home tools, or immersive environments, AwDS may encounter technologies that are not tailored to their preferences, contexts, or goals. Understanding how technologies have been designed for individuals with AwDS and how these individuals experience them is essential to ensuring more equitable and meaningful access.

In the United States, AwDS are living longer than ever before, with the mean and median age of death increasing from just 3 and 0 years in 1950 to 48 and 54 years in 2010 [40]. This transformation reflects decades of advocacy and reform in healthcare, education, employment, social services, and disability rights. As a result, people with **Down Syndrome (DS)** are increasingly participating in their communities, engaging in employment, continuing education, and building social relationships across their lifespan [66, 75, 133]. As their participation expands, so must our efforts to understand and support their interaction with technology, on their terms and in ways that reflect their identities, goals, and expertise. Historically, the perspectives and lived experiences of individuals with DS have often been overlooked in research and design, either by grouping them into broader disability categories or relying solely on caregiver or expert reports. Within research, the DS community is often grouped within the broader **Intellectual and Developmental Disabilities (IDD)** community [92]. Intellectual Disability is defined by the American Association on IDD as “a condition characterized by significant limitations in both intellectual functioning and adaptive behavior that originates before the age of 22” [113], where adaptive behaviors are the collection of conceptual, social, and practical skills that are learned and performed by people in their everyday lives [113]. Based on this definition, IDD may be non-specific or may encompass people from a diverse range of conditions and communities, including DS, **Autism Spectrum Disorder (ASD)**, Cerebral Palsy, Fragile X, Turner syndrome, Fetal Alcohol Spectrum Disorders, and so on. While there is overlap between the communities, each are distinct, and people with DS have unique and variable cognitive, physical, and social characteristics that warrant more specific attention in research and design [35, 46, 61, 117].

A growing body of systematic literature reviews has investigated how assistive technology can support individuals with IDD and DS, particularly in childhood. Boato et al. [19] reviewed virtual spaces for therapy and their effects on cognitive and motor development in children with DS. Their review identified games that supported motor coordination [18, 97], visual-motor integration [144, 152], and balance [5] through motion tracking and real-time feedback. They found that children often responded positively to technologies that included visual rewards, simple interfaces, play-based experiences, individualized calibration, and repetitive routines. Similarly, Shahid et al. [131] reviewed technology-enhanced support tools for children with DS and found that such technologies were used to support a range of abilities, including cognitive, social, physical, health, and autonomy-related skills. Their analysis highlighted tools that targeted cognition [2, 76], especially literacy

[65, 116], and autonomy [15, 45, 127]. Most applications and games support speech, numeracy, life skills, short-term memory, and emotions. They additionally identified AAC devices and **Tangible User Interfaces (TUI)** as studied technologies. They emphasized the value of tactile feedback and visual prompts and the importance of supporting multiple abilities. While their work included individuals aged 0–25 based on mental age classifications, the majority of reviewed tools were developed for early developmental use in classrooms or therapeutic contexts. This literature review removes papers focused on early development to identify areas of research being conducted with an older group.

Focusing on the educational or therapeutic setting, Krasniqi et al. [84] reviewed a range of immersive learning environments, mobile educational systems, AACs, and virtual simulations aimed at supporting independent skill acquisition. Their work included applications that guided users through daily tasks, literacy prompts, or created simulated environments for practicing functional routines. The review also identified persistent usability challenges limiting the integration into inclusive education, including inconsistent performance [58], difficulty with customization [72], a lack of connection across settings or mainstream technologies [87], and unclear interfaces [83]. These reviews focused on childhood development more related to classroom and therapy contexts. They did not assess how these design strategies translate into adult-centered goals such as community participation, employment, or independent living.

From a broader perspective, researchers have also investigated how technology can be developed in collaboration with people with IDD. Montoya-Rodríguez et al. [108] explored how **Virtual Reality (VR)** and **Augmented Reality (AR)** systems might be used to support adults with IDD, particularly in the development of social and communication skills. They reviewed interventions such as virtual job interview training [25, 26], educational games using avatars [28], and social scenario simulations [96]. Many of these systems provided immersive, engaging experiences that allowed participants to safely practice social behaviors, increase confidence, and learn cooperative strategies. Concluding VR/AR offers a rich, contextual learning environment that can be tailored to user needs. However, they also noted that few systems were developed specifically with or for individuals with DS, and most studies reported only short-term outcomes.

In a review of internet use among individuals with IDD, Glencross et al. [55] examined both the risks and benefits associated with digital participation. Their findings revealed that much of the existing literature identifies potential harms, such as online crime or exploitation [80, 95], emotional distress [31, 120], or internet overuse [94]. Despite these concerns, the review highlighted evidence that many people with IDD use the internet for social connection [17, 134], learning [80], independence and self-advocacy [120], and sexual expression [94, 95]. People with IDD reported feeling included and empowered when participating in online spaces, underscoring the importance of designing technologies that support safety, agency, and access.

Morash-Macneil et al. [109] conducted a systematic review focused on employment-related technologies for individuals with IDD. They found a growing trend toward using mainstream mobile technologies, such as smartphones and tablets, to support task sequencing, schedule reminders, navigation, and workplace communication. These tools offered advantages in terms of affordability, familiarity, and social acceptability [32], and many studies demonstrated improvements in job performance and independence. However, very few of the reviewed studies specifically reported on individuals with DS, making it difficult to assess how these benefits apply to that population.

Boot et al. [21] reviewed barriers to accessing assistive technologies for people with IDD across multiple contexts. They identified structural and systemic issues, including limited funding [129], inadequate caregiver training [38], and a lack of awareness [126]. The review recommended co-design and advocacy to enhance access and relevance. While they offered broad policy recommendations, their review did not focus on the unique life transitions and experiences of AwDS.

Despite the contributions of prior research, several gaps remain. Much of the literature focuses on children with DS, particularly in school or therapeutic environments, or groups AwDS and individuals with IDD together, limiting our understanding of the priorities and lived experiences of AwDS. Without more research in this area, we are unable to fully identify the distinctions or overlapping elements between communities. Adulthood also introduces new transitions and experiences. AwDS are increasingly taking on responsibilities related to employment, independent living, digital participation, and health management. To create assistive technologies that truly meet their needs, it is essential to understand what technologies exist and how they are studied, why they are applied, where they are used, and how AwDS themselves experience them. While assistive technologies can include both digital and non-digital tools, this review focuses on digital technologies used directly by an AwDS to assist them in performing a task or achieving a goal. We refer to “digital technologies” when discussing literature that examines digital inclusion (e.g., experience with mainstream technologies), as opposed to a specific assistive technology. “Digital and assistive technologies” and “technologies” will be used interchangeably to encompass both definitions.

Past reviews have categorized research methods broadly (e.g., “design studies,” “evaluation studies,” and “interviews”) without examining how accessibility and inclusion are embedded in the research process. This review seeks to clarify how studies have adapted their approaches to involve AwDS throughout the technology design lifecycle. We also examine the specific life areas technologies have targeted and the motivations behind these efforts. More specifically, we investigate whether there has been a shift away from educational assistive technologies that are commonly emphasized in child-focused research. This review also analyzes the benefits and challenges AwDS experience when using technology. While some findings may align with broader IDD research, our goal is to center the lived experiences of AwDS to inform more inclusive and relevant technology design. Guided by four **Research Questions (RQs)**, our goal is to map the current landscape of supportive technologies and inform future design efforts.

- RQ1. How have researchers involved AwDS in the design and development of technologies, and what inclusive methods or study approaches have been used?
- RQ2. In what ways has technology been identified as a means of supporting AwDS, and what motivations or perceived advantages have driven researchers to apply it in these contexts?
- RQ3. How have technologies been applied to support AwDS, and in which life domains are these applications most common?
- RQ4. What benefits and challenges have been reported in the use of technologies by AwDS?

2 Related Work

This section provides a critical background for situating the scope and focus of our systematic review. While the studies described here were not part of our formal review dataset, they offer important context for understanding the distinct experiences of AwDS and the research landscape surrounding them. We discuss the following: (1) DS in the broader disability landscape. We discuss how DS overlaps with other disability communities and conditions, underscoring the importance of recognizing shared and distinct support needs; (2) societal barriers in the U.S. for AwDS. We address the social, structural, and legal challenges that shape the daily experiences of AwDS, especially in contrast to children with DS. This highlights why an adult-focused review is both necessary and distinct; and (3) technological supports for children with DS outlines how past research has targeted younger age groups, helping contextualize the developmental shift in needs and the gap in adult-specific technology research that our review aims to address.

2.1 DS in the Broader Disability Landscape

DS is part of the broader category of IDD, a diverse group of conditions that affect cognitive function, adaptive skills, and overall development [6]. Many individuals with DS also have ASD (~10%–15%) [43], but it is often diagnosed later than in individuals with ASD alone due to overlapping behavioral traits and delays in motor and language development that complicate early identification [63, 122, 147]. Although individuals with DS and ASD share some developmental patterns with those with ASD only, they may present with poorer social communication and more stereotyped behaviors than peers with DS alone [56, 110]. Adults with DS-only tend to achieve higher levels of independence, social contact, and functional ability than their ASD-only peers [46].

Attention-Deficit Hyperactivity Disorder is also common among people with DS (~6%–30%) [43, 46] at a higher rate than the neurotypical population. However, ADHD may be underdiagnosed in this group, as symptoms often go unrecognized or are misattributed to general developmental delays. Children with DS and co-occurring ADHD frequently experience greater difficulties with inattention, hyperactivity, impulsivity, sleep, and behavioral regulation than peers with DS alone [44, 47]. These symptoms can be further complicated by medical conditions common in DS (e.g., hearing loss, visual impairments, and sleep apnea), which may mimic or exacerbate ADHD-related behaviors, making diagnosis and intervention more difficult [44]. Conductive hearing loss is common, with sensorineural hearing loss becoming more prevalent in adulthood, often leading to the use of hearing aids, cochlear implants, or sign language [6]. Visual impairments are also more frequent in individuals with DS, with many requiring glasses or surgical interventions. Alzheimer's disease significantly affects AwDS, with cognitive decline rates starting below 10% in individuals in their 40s, increasing to over 30% in their 50s, reaching 50%–80% for those aged 60–69 years, and 95% for 70+ [16, 118, 140]. Additionally, depression and anxiety are more common in individuals with DS compared to the general population [101], often related to communication difficulties and atypical symptom presentation. Mood disorders and anxiety disorders were found to be less common than those with IDD [10]. Psychological stressors, such as changes in living arrangements, school or care transitions, and bereavement, may also contribute to these mental health challenges [137]. Despite the prevalence of these comorbidities, **Human-Computer Interaction (HCI)** research is yet to extensively consider them when designing technology for individuals with DS. This may be due to the limited number of publications in the field. Acknowledging the complexity of DS and the variety of overlapping conditions it encompasses presents an opportunity to design technologies that potentially benefit a much broader population, while also highlighting the heterogeneity among AwDS. Many of the limitations in accessing technology in the DS community overlap with individuals with IDDs, acquired brain injuries, and aging adults. While the characteristics common for people with DS differ from those of other communities, overlapping needs, such as cognitive scaffolding, communication support, and sensory accommodations, are common across populations. For instance, individuals with traumatic brain injuries may also benefit from decluttered interfaces and simplified interface navigation to reduce cognitive load, much like AwDS [74, 156]. Aging adults may experience declines in memory, hearing, vision, or fine motor skills that mirror accessibility needs within the DS population [9, 23]. Aging AwDS also experience similar age-related declines earlier in life than neurotypical aging adults [73, 157]. The broader challenges of being stereotyped as incapable of using technology, experiencing poor interface design, or lacking access to personalized support apply across these communities. Future researchers should also consider collecting more detailed demographic data, with opt-out options, and investigate how overlapping conditions could influence technology access, ensuring that innovations are both inclusive and adaptable across different populations.

2.2 Societal Barriers in the United States of America for AwDS

Medical research has played a critical role in understanding DS, offering insights into its complexities. However, much of this research has historically framed disability through an impairment-based lens, focusing on deficits, leading to shifts in how we discuss disabilities [123]. The social model of disability provides a contrasting perspective, shifting the focus toward how societal structures and environments create disabling barriers [57]. Designing for individuals with DS should not be about “fixing” impairments but about reducing barriers to participation, autonomy, and inclusion. Legal frameworks, particularly guardianship laws, also limit autonomy for AwDS. Many people are placed under full or partial guardianship, which removes their legal right to make decisions about healthcare, finances, or relationships. While intended to protect, these arrangements often rely on outdated assumptions and overlook individual capabilities. Some adults with disabilities and their families prefer supported decision-making, which offers guidance without removing control [149]. However, the US legal system still relies heavily on guardianship, limiting personal independence, relationships, and financial autonomy [146]. Employment has a significant impact on the well-being and independence of individuals with AwDS, yet major barriers persist. Limited access to job-specific training and employment services, such as job search support, restricts skill development [86, 138]. Working memory challenges may further reduce employment chances [142]. Bush and Tassé [27] reported a decline in paid community employment, with only 16% of AwDS in such roles from 2011 to 2013. However, adults who regularly make daily decisions (e.g., how to spend their time or money) were more likely to be employed, highlighting the value of fostering decision-making skills. When informed about vocational opportunities, AwDS can make more empowered choices [60]. Still, many AwDS remain underemployed, working part-time or unpaid volunteer roles due to limited access to competitive, integrated positions and inadequate vocational skill development [33]. Housing and service access present additional challenges. Many AwDS continue to live with family well into adulthood, at a higher rate than peers with other IDD, largely due to long waitlists, high costs, and a lack of supported housing options [33]. Even with legal protections, community-based residential options remain limited, and families often provide care without sufficient professional support or respite services [24]. These issues are compounded for families from minority or immigrant backgrounds, who may encounter language barriers, cultural mismatches, or exclusion from planning processes [24, 64]. Some refugee families report feeling unheard or disrespected by professionals unfamiliar with their cultural norms around family roles, disability, or decision-making, leading to delays in support and misalignment in services [64]. Additionally, the lack of affordable or culturally responsive information systems makes it harder to navigate disability services or secure supports that reflect diverse needs. Fragmented services, limited specialist care, and insufficient funding for therapy services create additional hurdles for AwDS [33, 119]. The lack of appropriate programs and specialized professionals also limits participation in physical activities [100]. The absence of adequate programs for AwDS is a frequent barrier, as shown in the literature review by Souto et al. [138]. Leisure activities tend to occur primarily at home, with external barriers such as lack of nearby facilities, social connections, and transportation limiting broader community participation [106]. Barriers to public infrastructure and community organization services are facilitated by a lack of public disability programs and health services [138]. When compared to adults with ASD, Esbensen et al. [46] found that adults with ASD differed from AwDS across several domains of adult functioning. Adults with ASD had more difficulty completing daily living tasks and generally required more assistance, while AwDS were more likely to perform these tasks independently or with minimal support. Literacy levels also differed, with AwDS often able to read and write simple sentences, whereas adults with ASD typically could only manage a few words. Behavior problems were more prevalent among the ASD

group, who frequently exhibited multiple forms of maladaptive behavior, compared to fewer and less severe behaviors among AwDS. Finally, although both groups received similar numbers of services, AwDS appeared to benefit more from supports such as recreational programs, transportation, and speech therapy. Collectively, these findings highlight the structural, financial, cultural, and systemic barriers that shape the daily lives of AwDS and their families in the United States. However, this is not a comprehensive picture. Barriers may differ significantly in other countries or cultural contexts, where policies, services, and social attitudes toward disability vary widely. Continued research is needed to understand these intersecting influences across diverse global settings. Technology interventions have the potential to alleviate some of these barriers, enabling AwDS to access more opportunities and support.

2.3 Accessible and Assistive Technology for Children with DS

A broad range of technologies has been applied to address cognitive, communication, and motor abilities, tailored to the unique abilities of children with DS. More than 80% of children have started using computers by age six, mostly for learning and entertainment [50]. Many applications are designed to enhance cognitive skills, including speech [49, 71], numeracy [8], and health monitoring through mobile devices [82]. Games helped improve memory [2], tongue training [107], speech abilities [135], and emotional expression using avatars to teach children how to communicate emotions [67]. VR technologies have also been explored to enhance occupational therapy in children with DS to improve motor proficiency, visual-integrative abilities, and sensory integrative functions [41, 152]. Robots and mobile apps have been explored to provide support to therapists/teachers for designing therapeutic and educational intervention plans for children with DS [59, 125]. AAC devices were used to teach literacy skills and promote communication [12, 69]. Additionally, TUI, such as multitouch tabletops [42], foster the development of literacy and motor coordination [65, 76]. Feng et al. [51] highlight how different impairments impact technology use for children with DS. For example, fine motor limitations can cause input errors, but cognitive and language difficulties make it difficult to understand and recover from these errors without assistance. Children struggled with navigation, reading instructions, typing, and troubleshooting, and many lacked the patience or working memory capacity to follow unclear interface flows. While features like speech input or icon-based design might seem beneficial, the study warns that such solutions may backfire when other skill areas (e.g., verbal language or abstract reasoning) are also affected. Notably, children with DS demonstrate stronger social than cognitive or motor skills, which creates a mismatch as software that is cognitively accessible may feel too juvenile. At the same time, more age-appropriate programs often rely on unsupported abilities. These findings underscore the importance of design strategies that anticipate multichannel limitations, provide scaffolded support across age groups, and strike a balance between emotional engagement and cognitive demand.

3 Method

A systematic protocol was designed using the **Preferred Reporting Items for Systematic Reviews and Meta-Analysis Scoping Review (PRISMA-ScR)** guidelines, which support mapping the breadth and scope of research across varied study types [145]. While systematic in its search process and inclusion criteria, the review focused on exploratory synthesis and thematic coding rather than outcome-based analysis [37, 111]. Given the diversity of methods and limited emphasis on effect sizes or generalizable findings, a formal risk of bias assessment was not conducted. This review aims to answer the following RQs formulated using the **Population, Exposure, and Outcome (PEO)** question framework [103]. We selected the PEO framework because it is well-suited for reviews that include qualitative or mixed-methods studies, focusing on real-world

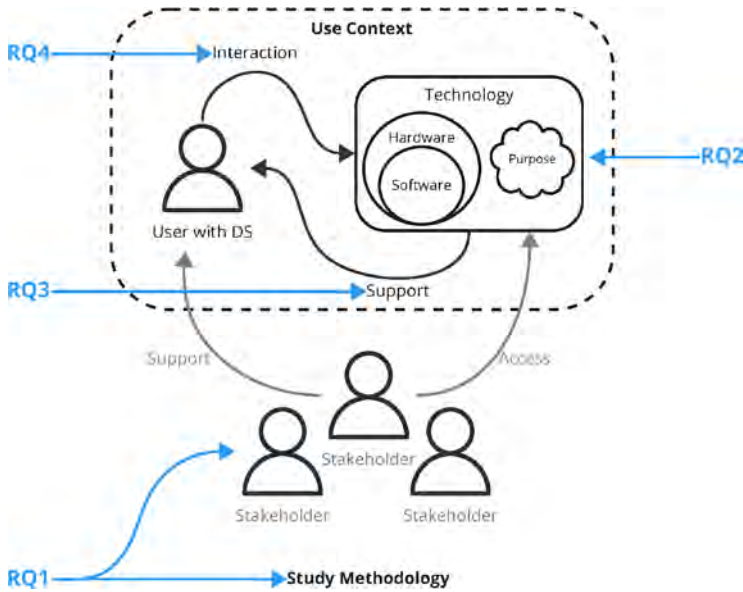


Fig. 1. Technology interaction and research methodology model. RQ1 focuses on the study methods and support systems employed in each study. RQ2 examines the technologies explored in these studies. RQ3 addresses the areas of support that technology provides to users with DS, while RQ4 explores the benefits and challenges associated with technology use for these individuals.

experiences and contextual factors. Unlike frameworks such as PICO, which are often tailored to clinical interventions or randomized controlled trials, PEO allows for a broader inclusion of studies. This framework informed our RQs, data attribute categories, and the criteria for inclusion and exclusion. Our questions were additionally derived from a technology interaction model we developed, shown in Figure 1.

RQ1. How have researchers involved AwDS (P) in the design and development of technologies (E), and what inclusive methods or study approaches have been used (O)?

RQ2. In what ways has technology (E) been identified as a means of supporting AwDS (P), and what motivations or perceived advantages have driven researchers to apply it in these contexts (O)?

RQ3. How have technologies (E) been applied to support AwDS (P), and in which life domains are these applications most common (O)?

RQ4. What benefits and challenges (O) have been reported in the use of technologies (E) by AwDS (P)?

For this review, we define “AwDS” as people who have DS and are 18 years old or older. We selected 18 as the cutoff because it is widely recognized as the legal age of adulthood in the United States and many other countries, with some exceptions, marking the point at which individuals typically assume legal responsibility for their decisions, actions, and personal autonomy [30, 48, 93]. In this review, we align our definition with standards used for the neurotypical population, recognizing and respecting the lived experiences and autonomy of AwDS. We define “assistive technology” as a digital tool used directly by a person with DS to assist them in performing a task or achieving a goal. We define “digital technologies” as mainstream technologies opposed to a

specific assistive technology. “Digital and assistive technologies” and “technologies” will be used interchangeably to encompass both definitions.

3.1 Systematic Review Method

In this review, we searched for publications in six databases: ACM Digital Library, IEEE Xplore, Compendex, Scopus, ScienceDirect, and SpringerLink. The ACM Digital Library, IEEE Xplore, and Compendex are technology-focused databases, whereas Scopus, ScienceDirect, and SpringerLink are all science-focused databases. Due to the multidisciplinary nature of the RQs, which involve both technology and the population, we selected a combination of technology- and science-focused databases. The databases were queried in November 2023.

3.2 Search String

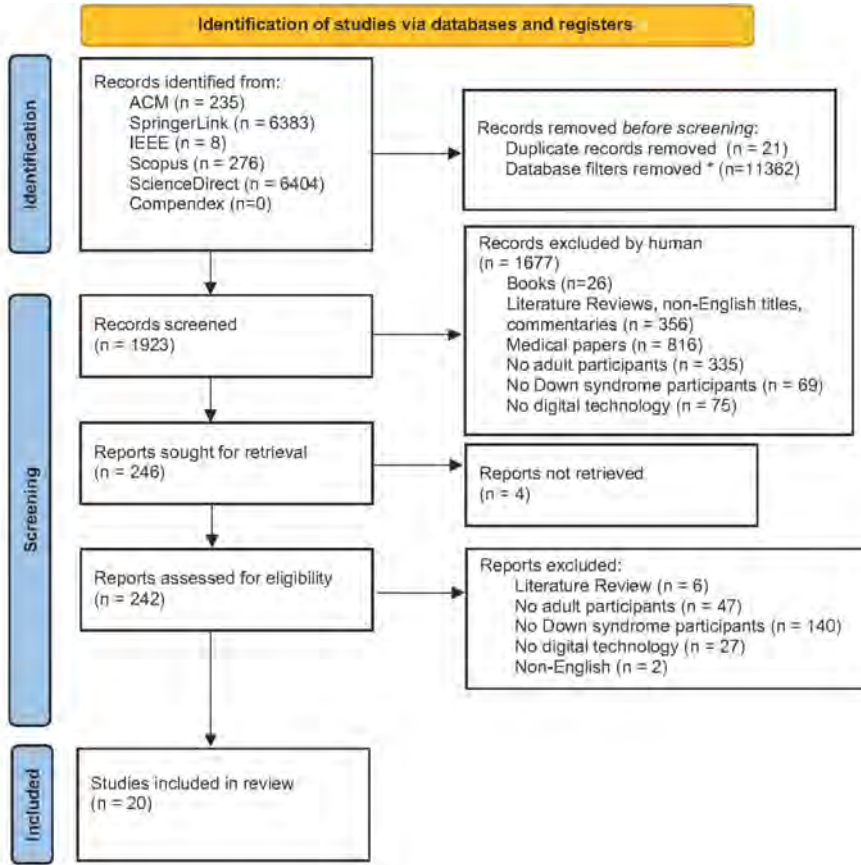
We searched all six databases with the search string “*Down syndrome*” AND *adult** AND ** technolog**, inspired by Shahid et al. [131] literature review. This search string contains three elements that all must be true to include a paper in the search results. The first element is the term “Down syndrome,” which forces both words in the term to be found together. This limits the chances of finding publications focused on other populations, such as Fragile X syndrome, Williams syndrome, Turner syndrome, and others. The second element is the term “adult*,” which includes an asterisk (i.e., wildcard search) at the end to accommodate for other forms of the word (e.g., adults, adulthood). The third element is the term “* technolog*,” where “technolog” was used with an asterisk before the term to retrieve results of various types of technology (e.g., assistive technology, accessible technology, support technology), and an asterisk at the end to accommodate for other forms of the word (e.g., technologies, technological). For databases that don’t allow wildcard searches (e.g., IEEE and ScienceDirect) because they automatically include searches with various forms of the three terms, we updated the search string to “*Down syndrome*” AND *adult* AND *technology*. No year limitation was made during the search process. All limitations are identified in Figure 2 and Table 1.

3.3 Inclusion/Exclusion

Our inclusion and exclusion criteria are outlined in Table 1. Publications were included if they: (1) involved participants with DS aged 18 or older, (2) focused on digital and assistive technologies directly used by AwDS, (3) reported on outcomes such as study methods, technology design, conceptual frameworks, technology usability and learnability, or areas of life that technology aimed to support, and (4) were published in peer-reviewed journals or conference proceedings. We excluded studies that focused on non-digital interventions, described only technical development without participant interaction, or did not specify participants as AwDS. Non-English articles, book chapters, abstracts, and duplicates were also excluded. Two independent reviewers participated in the screening process for this review. Titles and abstracts were screened separately by both reviewers using the inclusion and exclusion criteria. Articles that appeared to meet the criteria or required further evaluation were advanced to full-text review, with an interrater reliability of 0.78. Full-text articles were again assessed independently by both reviewers. Any discrepancies or conflicts in the inclusion and exclusion decisions were resolved through discussion and consensus, ensuring consistency and minimizing bias in the selection process. The final inclusion count was 20 publications, with an interrater reliability of 0.90.

3.4 Data Collection

Our data collection attributes and their definitions are presented in Table 2. Since our inclusion and exclusion criteria do not rely on the type of data collected, we investigated both qualitative and quantitative data. Each attribute was selected to address one of the four RQs. In cases where



* **ScienceDirect filters:** (1) English, (2) subject areas removed were "Medicine and Dentistry", "Biochemistry, Genetic and Molecular Biology", "Neuroscience", "Pharmacology, Toxicology, and Pharmaceutical Science", and "Immunology and Microbiology" (3) limited to "Research articles"

SpringerLink filters: (1) English, (2) Kept "psychology", "life sciences", "computer science", and "Science, Humanities and social sciences, multidisciplinary" and Removed "medicine and public health" and "biomedicine" (3) kept "conference papers" and "journal articles" (NOTE: SpringerLink was updated after paper identification and download. Therefore the current filters no longer fully align with the ones listed here)

Fig. 2. PRISMA publication identification flowchart.

a paper did not provide information on a particular attribute, that field was left blank in our data collection process and in Appendix A (Table A1), Appendix B (Table B1), and Appendix C (Table C1).

For each included paper in the literature review, we collected the corresponding attributes in Table 2. We collected metadata about each study, including the research discipline, publication venue, country of study, participant demographics, sample size, and the stated goal or purpose of the research. To minimize bias during the early stages of analysis, we extracted the remaining data attributes as verbatim quotes from the original publications, rather than paraphrasing or summarizing them. To address RQ1, we gathered information on the study's methods, including stakeholder involvement, data collection methods, study duration, data analysis approaches, and any notable strategies used to ensure accessibility or inclusion. To address RQ2, we extracted details

Table 1. Definitions of Chosen Inclusion and Exclusion Criteria

Criteria Group	Inclusion Description	Exclusion Description
Population	Participants were people with DS aged 18 and older	People with DS aged 17 and younger. Publications that do not include participants with DS
Exposure	Digital or assistive technologies (e.g., technology is a digital device that the user with DS uses directly, and it assists them in some capacity)	Non-digital technology solutions (e.g., paper instruction sheets, physical manufactured devices without digital elements)
Outcomes	Study methods, technology design, conceptual frameworks, technology usability and learnability, areas of support provided	Software or hardware development process (e.g., pure technical publications)
Type of publication	Focused on peer-reviewed journal articles and conference articles	Literature reviews, Book chapters, Commentaries, Posters, Abstracts
Date of publication	No limitation	No limitation
Language of publication	English	Non-English
Other		No PDF available, duplicate publications

The left column is the criteria group, following the PEO guidelines. The middle column is the description of the inclusion criteria. The right column is the description of the exclusion criteria.

about the technologies explored or developed, including their form (e.g., computer, smartphone application, smart home) and the rationale for their use. To address RQ3, we identified the area(s) of life the technology was intended to support (e.g., employment, communication, navigation), how the technology was used, and how effective or appropriate it was for the target population. For RQ4, we analyzed the reported benefits and challenges from both qualitative and quantitative findings, as well as key insights from the discussion and future work sections. Finally, we identified design recommendations based on a mixture of RQ3 and RQ4 data attributes. After collecting all attributes, we conducted a qualitative coding process on the extracted quotes, which informed the thematic groupings and categorizations presented in Section 5.

3.5 Data Analysis

Following data collection for each attribute listed in Table 3, we employed a combination of qualitative analysis and structured categorization to address the four RQs. Our analytic process was guided by Clarke and Braun’s six-phase approach to thematic analysis [37], but applied differently depending on the nature of each RQ.

For all RQs and design recommendations, we began with the first three phases: (1) familiarizing ourselves with the data by reading each paper in full, extracting quotes relevant to each data attribute, and reviewing the extracted quotes; (2) generating initial codes by identifying recurring ideas, patterns, or functions in the quoted material; and (3) grouping the data into preliminary categories based on conceptual similarity and relevance to the RQs. This approach enabled us to maintain adherence to the language and framing employed by the original authors while identifying meaningful patterns in how technologies were designed and applied.

For RQ1 through RQ3, we used these preliminary groupings to develop structured categorizations rather than full themes. This decision reflected the nature of the data, as many of the extracted quotes for these questions described specific hardware, software types, technology applications, or

Table 2. Definitions of the Selected Data Attributes Gathered during the Data Collection Process

Data Attribute	Attribute Description	Collection Purpose
PID	Unique ID assigned to each paper	Identification of papers
Paper type	Classifies type of publication (e.g., conference, journal)	Required by the inclusion criteria
Field of focus	Disciplinary field of the research	Understand which fields are engaging in this research
Publishing platform	Name of publishing venue and location	Identify common venues for dissemination
Country of research	Country where research was conducted; if not specified, use first author's affiliation	Understand geographic distribution of research
Inclusion/Exclusion	Boolean: True = included; False = excluded	PRISMA-ScR methods
Reason for exclusion	Explanation of exclusion if applicable	Transparency in screening process
Goal of paper	Stated aim or RQs of the paper	Identify gaps in research
Participant demographics	Participant demographic breakdown, if specified	Identify gaps in population
Sample size	Total number of AwDS participants	Identify population size and representation
Stakeholder support	Description of caregiver, educator, or other stakeholders, if applicable	RQ1 data
Data collection methods	Type of data collected and study design	RQ1 data
Study time duration	Number of sessions, session length, task quantity, or timeline	RQ1 data
Data analysis techniques	Analysis used to inform results	RQ1 data
Inclusive methodology information	Any notable methodology details related to accessibility	RQ1 data
Technology developed or studied	Type of technology developed or studied (e.g., app, AR)	RQ2 data
Technology specifics	More detailed software/hardware details	RQ2 data
Technology use rationale	Description or theory explaining why the technology was used	RQ2 data
Area of support	Life domain(s) the technology supported (e.g., employment)	RQ3 data
Technology intervention	How the technology was used during the study	RQ3 data
Technology effectiveness	Reported success or failure of technology for participants	RQ3 data and design recommendations
Key quantitative results	Outcomes from statistical analyses	RQ4 data and design recommendations
Key qualitative results	Themes, quotes, or participant experiences	RQ4 data and design recommendations
Discussion/future work	Authors' reflections, limitations, next steps	RQ4 data and future work

Table 3. Identified Benefits of Technology Use across Studies

Citation	Confidence and Engagement	Independence and Digital Inclusion	Skill Development	Communication	Social Inclusion and Connection
Kumin et al. [85]	-	x	-	x	x
Alemi and Bahramipour [4]	x	-	x	x	-
Lazar et al. [92]	x	x	-	-	x
Bargagna et al. [13]	x	-	x	-	-
Lussier-Desrochers et al. [98]	x	x	-	x	x
Holyfield et al. [70]	x	-	x	-	-
Mechling et al. [104]	-	-	x	-	-
Wilkinson and Bennett [150]	-	-	-	x	x
Holyfield and Drager [68]	-	-	-	x	-
Ma et al. [99]	-	x	x	-	-
Buzzi et al. [29]	x	-	x	-	-
Dalprà et al. [39]	x	x	x	-	-
Gallaher et al. [54]	x	x	x	x	-
Villamil Matallana and Paredes-Velasco [148]	x	-	x	-	-
Jevne et al. [77]	x	x	-	x	x
Khan et al. [79]	-	x	-	x	x
Lazar et al. [91]	-	x	-	x	x
Lachapelle et al. [88]	x	x	-	-	x
N’Kaoua et al. [89]	-	x	x	-	-
Boleracki et al. [20]	-	-	-	-	-
Total	11	11	10	9	8

“x” = benefit documented; “-” = not documented.

supported life domains. Accordingly, we organized the findings by technology type, application purpose (e.g., learning, support, or practice), and the areas of life being addressed (e.g., communication, independent living, navigation). These groupings were descriptive and comparative rather than interpretive. For RQ4 and the design recommendations, we continued with the full thematic analysis process. After coding the quotes related to benefits, challenges, and design considerations, we (4) reviewed and refined themes to ensure internal consistency and clear conceptual boundaries; (5) defined and named the resulting themes to reflect their contributions to understanding user experience and inclusive technology design; and (6) synthesized these themes into a structured narrative. All four RQs and design recommendations informed our final discussion and future directions.

4 Metadata Results

This section presents the metadata analysis, which covers: (1) an overview of the inclusion process; (2) demographic information; (3) distributions of publications by year and region; and (4) the academic disciplines represented in the studies. This section provides context for the findings presented in Section 5.

4.1 Included Publications

After combining results from the six identified databases, we compiled a total of 1,944 publications (ACM = 235, SpringerLink = 1,026, IEEE = 8, Scopus = 276, ScienceDirect = 399, Compendex = 0). We removed 21 duplicate publications, for a total of 1,923 screened. The exclusion process is detailed

in Figure 2. We began with a title and abstract screening, only excluding publications if we were certain they did not meet our inclusion criteria. In cases of uncertainty, we retained the publication for full-paper screening. After removing books, commentaries, non-English publications, literature reviews, medical publications (e.g., neonatal, diagnosis, motherhood), publications focused on individuals under 18 years old, studies not involving people with DS, and publications misaligned with our definition of technology, we were left with 246 publications for full-paper screening. During full-paper screening, we excluded 6 literature reviews, 47 publications that included people under 18, 140 studies that did not focus on people with DS, 27 publications misaligned with our definition of technology, 4 publications that were inaccessible, and 2 that were non-English publications. Ultimately, 20 publications were included in this literature review.

4.2 Demographic Breakdown

Across the 20 publications identified, the average age of AwDS was approximately 24.4 years, with most studies including participants between 18 and 30 years old. The youngest participant was 18 years old, and the oldest was 44. The gender distribution across studies was relatively balanced, with 2.94 males per study ($SD = 2.53$) and 2.89 females per study ($SD = 2.56$). Sample sizes of AwDS ranged from 1 to 15 participants per study, with an average of about 6.5. While some studies reported detailed demographic breakdowns, including age ranges and gender, others provided only limited or no demographic data. In a few cases, studies also included stakeholder groups as participants or comparison groups (e.g., typically developing participants as controls, caregivers, educators); however, we did not include their demographic breakdown in this summary in Appendix B. Overall, while the demographic data provides useful insight into young adult experiences with technology, it also highlights the need for broader inclusion of older adults and larger, more representative samples. These gaps may also reflect systemic challenges in accessing older participants in community settings or engaging those with more complex support needs.

4.3 Year and Region Distributions

We collected data on the year of publication and the country where the research was conducted, as shown in Figure 3. Country designation was determined based on the location from which participants were recruited; however, if participant location was not specified, we used the first author's country as a proxy. The earliest identified publication was in 2002, followed by an eight-year gap with no recorded studies. Since 2011, a consistent, albeit small, increase in publications in this area has been observed. Notably, there were no publications in 2017 or 2020, with the COVID-19 pandemic disruption likely contributing to the absence of publications in 2020 [102]. These trends suggest that the development of assistive or accessible technologies for AwDS is a relatively recent research focus that is gaining traction at a steady pace.

The distribution of research across countries was categorized according to the United Nations geoscheme for geographic regions.¹ Based on continental groupings, the majority of publications (11 out of 20, or 55%) originated from the Americas, with eight studies conducted in the United States and two in Canada. Europe accounted for the second-largest share, with seven publications (35%), including two from Italy and one each from Spain, Norway, Scotland, France, and Hungary. Asia was represented by a single study from Iran (5%), and Oceania by one study from Australia (5%). No publications were identified from Africa, Asia (except for Iran), South America, the Caribbean, Eastern Europe (excluding Hungary), and Pacific Island nations. This distribution highlights a strong skew toward research conducted in Western countries, particularly the United States and Europe. As a result, the findings of this review may reflect cultural norms, values, and support

¹<https://unstats.un.org/unsd/methodology/m49/>.

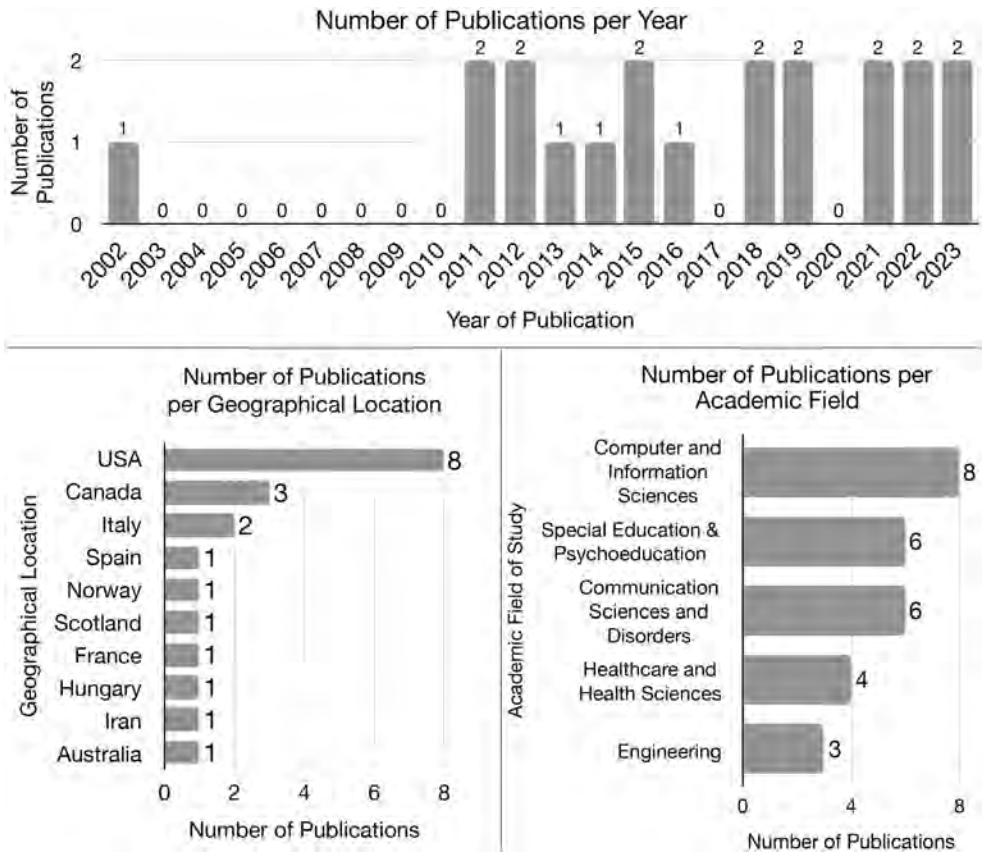


Fig. 3. *Top*: Bar chart showing the number of publications per year. *Bottom left*: Bar chart showing the number of publications per continental region. *Bottom right*: Bar chart showing the number of publications in each academic field of study. Nine publications featured multidisciplinary author teams, while the remaining 11 had authors from a single academic department. Two publications included community groups as contributing authors.

systems specific to these regions. Future research should prioritize expanding geographic diversity and examining how cultural and regional differences influence the goals, preferences, and lived experiences of AwDS. Broader representation is crucial for developing inclusive and culturally responsive technologies that cater to the diverse needs of global populations.

4.4 Fields of Publication

We collected data on the affiliated departments of authors across all 20 publications, as shown in Figure 3. Of these, 9 featured multidisciplinary author teams, while the remaining 11 had authors from a single academic department, as shown in Appendix A. Additionally, two publications included community groups as contributing authors. The fields represented were categorized as follows: (1) Computer and Information Sciences ($n = 8$, 40%); (2) Special Education and Psychoeducation ($n = 6$, 30%); (3) Communication Sciences and Disorders ($n = 6$, 30%); (4) Healthcare and Health Sciences ($n = 4$, 20%); and (5) Engineering ($n = 3$, 15%). This disciplinary diversity highlights a growing recognition of the importance of designing technologies for AwDS. Each field brings

distinct perspectives and expertise to the design of inclusive systems that can better align with the needs, strengths, and contexts of AwDS.

5 RQs Results

This section presents the results and key findings based on our analysis: (1) examine the common methodological approaches and inclusive strategies used to study technology with AwDS; (2) explore the technologies investigated and the motivations behind their use; (3) discuss the purposes these technologies served and the areas of life they were intended to support; (4) summarize the reported benefits and challenges experienced by AwDS when using these technologies; and (5) present design recommendations for future researchers and developers.

5.1 RQ1. How Have Researchers Involved AwDS in the Design and Development of Technologies, and What Inclusive Methods or Study Approaches Have Been Used?

We have identified five categories of study methods that researchers have employed to study technologies with AwDS: (1) experimental studies; (2) usability studies; (3) exploratory studies; (4) participatory design studies; and (5) observational and ethnographic studies. Since these studies are conducted with a vulnerable population, extra care was taken to ensure respect for the participants and their personal agency. Based on our analysis, researchers considered six categories of inclusive strategies when working with the DS community: (1) stakeholder or expert involvement; (2) study customization and personalization; (3) support in study instructions and task completion; (4) comfortable environment adaptability; (5) visual study aids; and (6) preparation with the technology or extra time.

5.1.1 Study Methods. The publications included in this review employed a range of methods to explore various aspects of technology use for AwDS. We identified five key categories of methods: (1) experimental studies ($n = 6$); (2) usability studies ($n = 5$); (3) exploratory studies ($n = 4$); (4) participatory design studies ($n = 3$); and (5) observational and ethnographic studies ($n = 2$). These categories reflect the diversity of research approaches. In addition to these methodological categories, five of the studies were pilot studies, and four were case studies, due to the relatively early and exploratory nature of this research area.

Experimental studies ($n = 6$) were the most commonly used method, which aligns with the interdisciplinary nature of the research topic. These studies focused primarily on testing specific technology interventions or hypotheses under controlled conditions. These included single-subject multiple-probe designs [70, 104], between-groups comparisons [89], and pre/post testing [4, 54]. The studies commonly measured quantifiable outcomes such as task accuracy, communication behaviors, or learning gains. For instance, Mechling et al. [104] used video modeling to support the completion of fine and gross motor tasks, measuring participants' task performance as an outcome. Holyfield et al. [70] tested a literacy-focused app, systematically analyzing changes in word decoding accuracy across multiple sessions. Similarly, Wilkinson and Bennett [150] examined communication performance as participants used different types of AAC displays. One study incorporated comparative designs to evaluate technological effectiveness across groups. N'Kaoua et al. [89] conducted a between-group comparison of navigational skills among AwDS, typically developing children, and adults, measuring participants' wayfinding accuracy and behavior in controlled settings. In addition, two studies described as case studies also featured experimental components. Alemi and Bahramipour [4] explored English vocabulary acquisition using a robotic teaching assistant, tracking learning gains over time. Gallaher et al. [54] assessed the effectiveness of a computer-assisted literacy program with a single participant, measuring reading improvement

over time. These experimental studies provided structured, repeatable evidence on performance impacts but generally offered limited insight into longer-term use or personal experiences.

Usability studies ($n = 5$) assessed the practicality of technology and the effectiveness with which participants use it in various settings. Usability studies commonly evaluate a system's effectiveness, efficiency, and user satisfaction in a given context [1]. For example, Kumin et al. [85] implemented a task-based usability evaluation of workplace-relevant applications on a tablet, while Ma et al. [99] conducted a comparative usability study of different password authentication systems between AwDS and neurotypical individuals. They collected data on user performance, preferences, reasons for failure, and password strength. Three studies were also framed as pilot studies [13, 29, 88], emphasizing early-stage testing of prototypes or platforms. Bargagna et al. [13] aimed to enhance the functionalities and user interfaces of a telerehabilitation platform developed in collaboration with psychologists and doctors. Similarly, Buzzi et al. [29] conducted formative user testing on a telerehabilitation tool, focusing on initial feedback and potential cognitive benefits. Lachapelle et al. [88] assessed a navigation app with a single participant, using standardized instruments and post-test interviews to evaluate both usability and user satisfaction in a real-world context. Although some studies included multiple conditions or phases, they remained grounded in the user-centered goals of system evaluation, rather than intervention testing.

Exploratory studies ($n = 4$) were used to investigate emerging questions or underexplored areas related to technology use by AwDS, often using qualitative interviews, descriptive observations, or prototypes rather than outcome-based testing. Jevne et al. [77] conducted interviews to explore the subjective well-being of emerging AwDS, capturing personal reflections and life experiences in relation to technology. Holyfield and Drager [68] examined the feasibility of integrating speech recognition into AAC devices by assessing the intelligibility and consistency of speech produced by AwDS. Their study compared accuracy in understanding AwDS across familiar listeners, unfamiliar listeners, and standard speech recognition systems. Villamil Matallana and Paredes-Velasco [148] conducted a case study that evaluated a dance teaching system used in real-world educational settings. They combined quantitative questionnaires, observations, and caregiver reports, focusing on experience and teaching context. Lussier-Desrochers et al. [98] conducted a pilot study with longitudinal qualitative interviews over 10 months to investigate the digital inclusion of AwDS across multiple domains, including communication, commuting, entertainment, home life, and employment. This study incorporated participatory design in its framing; however, the participants were not involved in the decision-making process, so we categorized it as exploratory. foundation for more structured research approaches, including participatory design studies. These studies often surfaced insights into contextual barriers, technology experiences, and everyday routines, providing critical input for subsequent design phases.

Participatory studies ($n = 3$) involve participants' perspectives and opinions during the design and development process. This approach ensures that the technologies created are shaped by the users' needs, preferences, and lived experiences, making the resulting tools more accessible and relevant. For instance, Khan et al. [79] utilized participatory action research methods to develop mobile applications that support independent living. They worked closely with participants to understand their requirements throughout multiple design cycles. Lazar et al. [92] conducted a case study to co-design an application that supports nutritional decision-making for AwDS in their daily lives. By involving participants in the design process, these studies produce tools that are more in line with the specific abilities and preferences of their intended users.

Observational and ethnographic studies ($n = 2$) captured behavior and use patterns in natural or semi-controlled environments, without manipulating interventions or tightly structuring tasks. For example, Lazar et al. [91] conducted an ethnographic study to observe the computer skills of AwDS in professional settings, providing insights into how these individuals adapt to and engage with

workplace technologies. Similarly, Dalprà et al. [39] conducted a longitudinal pilot study in which participants resided in their research apartment for 4 consecutive days per month over a total period of 22 months, spanning 3 years. They observed how a smart home system affected the daily living skills of an AwDS, tracking their behavior in the home environment. These observational studies provide critical insights into how AwDS utilize technology in their daily lives. Although some exploratory studies incorporated informal observations, we classified only those that used observation or ethnography as a primary method, not simply as a supporting technique, under the “observational and ethnographic studies” category.

Together, these methodological approaches offer a comprehensive understanding of how technologies can be effectively designed, tested, and implemented to support the heterogeneity of AwDS. Each method brings distinct strengths. Experimental studies generate measurable data on intervention effectiveness; usability studies evaluate practical system performance in real-world tasks; exploratory studies surface emerging needs and assess feasibility using diverse data sources; participatory studies center the lived experience of AwDS through co-design and collaboration; and observational and ethnographic studies uniquely capture how technologies are embedded into everyday routines through sustained, naturalistic observation. While some categories contained only a few studies, this more granular breakdown enables clearer analysis of how the field approaches technology development and evaluation for AwDS, highlighting areas where certain methods remain underutilized.

5.1.2 Data Collection. Different data collection methods were employed across the various study designs. Figure 4 illustrates the relationship between study methods and chosen data collection techniques. Qualitative data collection was the most frequently employed ($n = 7$) [13, 77, 79, 88, 91, 92, 98], followed by quantitative methods ($n = 6$) [4, 68, 89, 99, 104, 150] and mixed-methods approaches, which combined qualitative and quantitative techniques ($n = 6$) [29, 39, 54, 70, 85, 148]. One publication was a proposal paper without a specified data collection method [20]. The prominence of qualitative methods reflects the importance of capturing participants’ lived experiences, preferences, and perspectives. These approaches enable researchers to examine the intricacies of user interaction with technology, encompassing motivation, satisfaction, and contextual challenges. In contrast, quantitative methods provide structured, measurable outcomes, such as task accuracy, completion rates, or time on task, which are often used to evaluate the effectiveness or usability of specific technologies in controlled settings. Mixed-methods approaches leveraged the strengths of both paradigms, offering a more comprehensive understanding by combining statistical results with user-reported experiences.

5.1.3 Approaches to Creating Inclusive Study Environments. Researchers reported various approaches they employed to ensure their study environment was inclusive of their participants. These approaches foster an environment conducive to effective participation by considering and respecting each participant’s abilities. Six categories of inclusive strategies were employed across studies: (1) stakeholder or expert involvement ($n = 11$); (2) study customization and personalization ($n = 6$); (3) support in study instructions and task completion ($n = 6$); (4) comfortable environment adaptability ($n = 5$); (5) visual study aids ($n = 5$); and (6) preparation with the technology or extra time ($n = 3$). Some of these approaches may have been used in more studies than explicitly reported. Five publications didn’t report utilizing any of the categorized inclusive methods. Between one and four inclusive approaches were the most common choice, occurring in four publications each. Two publications employed six out of the seven approaches. As you can see in Figure 4, each method category described in Section 5.1.1 is adaptable to incorporate inclusive approaches.

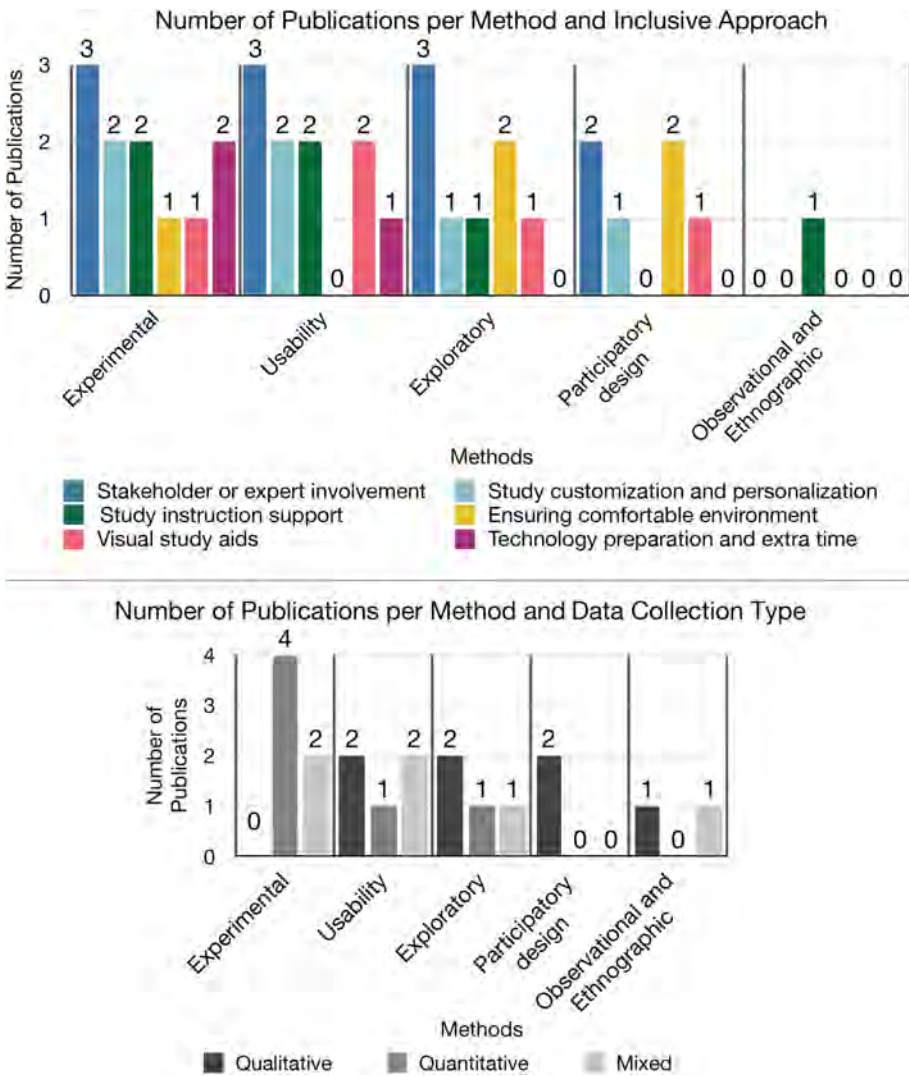


Fig. 4. *Top*: The number of publications per study method category and approaches to creating inclusive study environments. Five publications didn't report utilizing any of the categorized inclusive methods. Between one and four inclusive approaches were the most common choice, occurring in four publications each. Two publications employed six out of the seven approaches. *Bottom*: The number of publications per study methods category and data collection type. Each publication was assigned one data collection type.

Stakeholder or expert involvement ($n = 11$) was the most prevalent inclusive approach. Stakeholders, including parents, caregivers, educators, and therapists, offered feedback, provided varying viewpoints, and ensured participant support throughout the study. For instance, Alemi and Bahramipour [4] involved teachers within lessons. Similarly, teachers and parents provided feedback in an education study [148], and a teacher ran the literacy program with feedback from the participants' parent [54]. Other experts, including dietitians and caregivers, were also involved to enhance a nutrition-tracking app [92]. Khan et al. [79] conducted focus groups with caregivers to help develop independent travel apps. Holyfield and Drager [68] included familiar listeners to

assess speech intelligibility. Wilkinson and Bennett [150] assessed social interaction in structured scenarios. Jevne et al. [77] allowed participants to choose if they wanted a parent present during the interviews (with seven out of eight AwDS choosing to include a parent). And Lachapelle et al. [88] included the perspectives of parents and caregivers on the technology's usability.

Study customization and personalization ($n = 6$) tailored study tasks or materials to align with participant preferences. These adaptations helped ensure that participants could fully engage with the study activities and that data collection reflected their authentic experiences. In Jevne et al. [77], participants were invited to bring in personal photos, which were used to guide interview conversations and make the process more meaningful. The semi-structured interviews were adapted to fit each participant, allowing for greater flexibility during the sessions. Kumin et al. [85] allowed flexibility in task content, enabling participants to type different words or input different pieces of data during the study tasks. Lazar et al. [92] provided personalized avatar illustrations that reflected each participant's hobbies to represent participants during focus group sessions. Lachapelle et al. [88] modified the study tasks to align with each participant's interests, making the navigation activities more relevant and engaging. Finally, Wilkinson and Bennett [150] began their training program by assessing each participant's individual goals, which were then used to tailor the program's content.

Supporting study instructions ($n = 6$) involved researchers providing additional guidance or scaffolding to help participants understand tasks and engage meaningfully with the study. In Kumin et al. [85], instructions were presented both in written form and read aloud before each task, providing participants with multiple modes of understanding. Similarly, Buzzi et al. [29] described the platform to participants and walked them through its main functionalities, ensuring they were comfortable with the interface before beginning the study. In Mechling et al. [104], verbal and gestural prompts were used to guide participants when they made a mistake during a task, helping to maintain engagement and task flow. Chapman and Mitchell [34] implemented a structured support system that included four levels of assistance: verbal intervention, articulated verbal intervention, operational action, and full physical assistance, depending on the participant's level of need during each task. Holyfield and Drager [68] read each sentence aloud to support the participant in saying the sentence they heard to record for speech collection. And Wilkinson and Bennett [150] utilized a storybook to study social communication.

Ensuring a comfortable environment ($n = 5$) for participants was implemented by conducting studies in familiar or participant-preferred settings and making thoughtful adjustments. In Alemi and Bahramipour [4], the study was conducted in a classroom setting, allowing participants to engage with the technology in a familiar educational environment. They also conducted warm-ups through songs or games. Holyfield and Drager [68] conducted their study in participants' homes and used tablet microphones instead of head-mounted devices to avoid discomfort and make the experience more natural. Similarly, Khan et al. [79] also conducted sessions in participants' homes to support ease of participation. Jevne et al. [77] offered flexible study locations, including participants' homes, respite homes, workplaces, and even cafes, based on where individuals felt most comfortable. Lazar et al. [92] took additional steps to build rapport and comfort by giving participants time to get to know each other and their surroundings, and by learning about their personal interests and hobbies prior to focus group sessions.

Visual study aids ($n = 5$) were incorporated to enhance accessibility for participants who had difficulty with reading or abstract verbal instructions. These aids were object-based and static, compared to the prior category of supporting study instructions, which was dynamic and process-oriented. These visual supports helped clarify tasks, prompt responses, and guide participants through the study process in a more intuitive and engaging way. In Kumin et al. [85], visual Likert scales were employed to facilitate participant feedback and ensure that responses could be easily

understood and selected. Alemi and Bahramipour [4] used flashcards with images of real objects as teaching tools. Lazar et al. [92] storyboarded the flow of the study session to visually communicate what would happen next, making the structure of participation more accessible and predictable. Holyfield and Drager [68] used a tablet-based interface that displayed images to prompt participants to say specific words as part of a speech recognition assessment. In Buzzi et al. [29], participants were provided with visual demonstrations for each game, showing how the interface worked before they interacted with it themselves.

Preparation with the technology and extra time ($n = 3$) helped participants feel more comfortable and confident before engaging in the main study tasks. Comfort and confidence were key goals in this category, as they were in ensuring a comfortable environment; however, this category focuses on device and task familiarity, as opposed to the physical and social context. These preparatory steps often involved warm-ups or guided introductions to the devices or systems being used. In Kumin et al. [85], a warm-up task was added to help participants become familiar with both the researchers and the tablet interface before beginning the usability activities. N’Kaoua et al. [89] implemented a training phase in which participants explored the virtual environment and practiced using the joystick to navigate, ensuring they understood the controls before formal data collection began. Wilkinson and Bennett [150] allowed participants to choose between one and two sessions to complete the entire study.

Incorporating inclusive study approaches ensures that researchers are supporting a wider range of participants. While these strategies often share common goals, such as promoting comfort, clarity, and engagement, we have separated them into six categories based on their distinct focus. Stakeholder involvement emphasizes who is engaged, such as caregivers and educators, while study customization addresses what is tailored to participant needs. Instructional support refers to the delivery of information in real-time, whereas visual aids are static materials that enhance clarity. Ensuring a comfortable environment relates to the physical and social setting, and preparation time helps participants build confidence with technology. Distinguishing these categories highlights the multifaceted nature of inclusive research and provides concrete strategies across study design and execution.

5.2 RQ2: In What Ways Has Technology Been Identified as a Means of Supporting AwDS, and What Motivations or Perceived Advantages Have Driven Researchers to Apply It in These Contexts?

To answer our second RQ, we began by identifying the technologies being investigated across the included studies and then analyzed the researchers’ reasoning behind their use of technology. To support this analysis, we identified the technologies described in each paper using direct quotes, then grouped the studies based on the primary hardware platform used (e.g., computer, tablet, smartphone). Within each hardware category, we further classified the technologies by their associated software or functionality to reflect how the hardware was used in context, as shown in Figure 5.

5.2.1 Technologies Studied. Across the 20 studies reviewed, a range of technologies were explored to support AwDS, with notable variation in hardware platforms, software types, and intended functions. A substantial number of studies focused on *computers and computer-based web technologies*. These included workplace and productivity tools, such as e-mail, word processors, and browsers, for developing digital literacy and employment readiness [91], as well as literacy training programs utilizing phonics software and internet services [54]. Web platforms were commonly used to deliver structured interventions, ranging from telerehabilitation for cognitive training [13], to gamified learning environments with AAC features [29], to simulated e-commerce platforms for

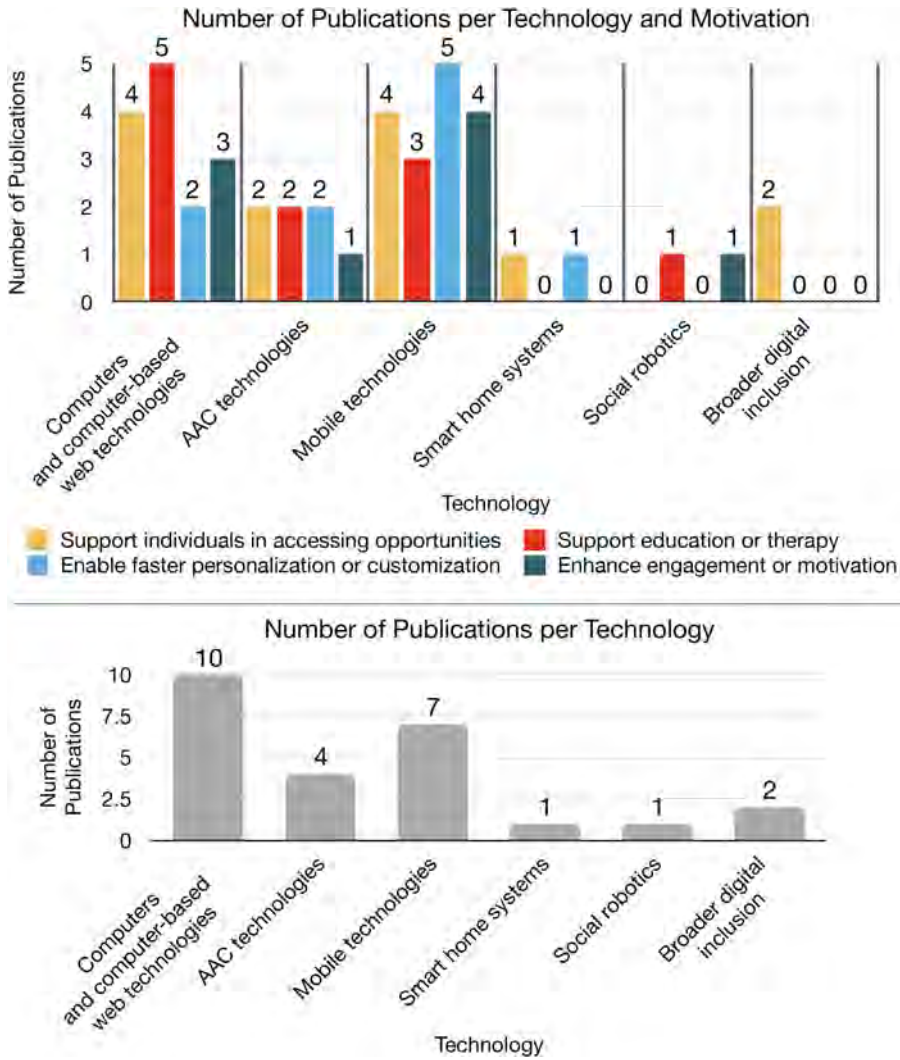


Fig. 5. *Top*: The number of publications per technology category and researchers' motivation to use technology. Eleven publications have been assigned more than one category. Support education or therapy/enhance engagement or motivation ($n = 4$); support accessing opportunities/faster personalization or customization ($n = 3$); support accessing opportunities/enhance engagement or motivation/faster personalization or customization ($n = 2$); support education or therapy/enhance engagement or motivation/faster personalization or customization ($n = 1$); support education or therapy/faster personalization or customization ($n = 1$). *Bottom*: The number of publications per technology category. AAC technologies are highlighted in this figure as a separate category, despite all being hosted on computers, resulting in 14 publications within computers and computer-based web technologies. One publication was assigned to both computers and computer-based web technologies and mobile technologies.

testing authentication systems such as mnemonic and graphical passwords [99]. Several studies employed computers as delivery systems for immersive or multimedia-based training, including video modeling via PowerPoint [104], and virtual environments for practicing navigation [89] and independent living skills [20]. AAC technologies were also studied through desktop interfaces,

including customized storybook-linked symbol displays [150], decoding supports with dynamic text and visual scene displays [70], and the potential integration of speech recognition software like Google Speech-to-Text for AAC access [68].

Mobile technologies were also prominently featured, especially for real-time, context-aware interventions. Smartphone-focused studies included tools for nutrition decision-making in restaurants [92], navigation with GPS tracking and caregiver alerts [79], and task guidance using simplified interface overlays with picture and audio prompts [88]. One tablet-focused study included those that evaluated workplace productivity task completion on iPads [85]. Some studies have leveraged both smartphones and tablets as delivery platforms, such as the MyDanceDown application, which enables dance-based learning and feedback both in school and at home, using teacher-generated, personalized videos [148]. These mobile technologies were often chosen for their portability, personalizability, and familiarity, supporting use in diverse life contexts such as travel, leisure, learning, and communication.

Other studies assessed *smart home systems*, *social robotics*, and *broader digital inclusion*. A fully automated training apartment was evaluated for its ability to support daily living skills, including cooking, cleaning, and schedule management, through touchscreens and mobile devices [39]. A humanoid robot, NAO/NIMA, was studied as a teaching assistant in a language-learning program, supporting vocabulary acquisition through interactive engagement [4]. Finally, two studies were conceptually focused, with one examining how digital inclusion strategies could enhance participation across life domains for AwDS, without evaluating a specific tool [98], and another exploring the subjective experiences of well-being and quality of life among emerging AwDS [77]. These papers provide valuable context for the broader technological landscape and user needs, even though no specific systems were implemented.

5.2.2 Motivation to Use Technology. Researchers chose to explore various technologies for several reasons. The four identified categories outline the reason alone. We are not claiming that each paper successfully achieved its goal within the studied technologies or that these categories are an exhaustive list of why technology might be developed for AwDS. The four key categories identified for employing technology were: (1) to support individuals in accessing online or community opportunities ($n = 11$); (2) to support education or therapy, alongside a teacher or therapist, or at home ($n = 8$); (3) to enable faster personalization or customization, allowing technologies to be tailored to individual user preferences ($n = 7$); and (4) to enhance engagement or motivation while completing tasks ($n = 7$). In one paper, technology was being used to assess the spatial skills of AwDS [89].

Supporting individuals in accessing opportunities was, unsurprisingly, the most common reasoning for using technology, whether that be employment, social situations, digital inclusion, independent living, or independent navigation. In a broad sense, it is possible that all 20 publications utilized technology to achieve this goal; however, some publications didn't emphasize or state this as their reasoning, so we did not include them in this category. Technology was chosen to provide workplace assistance to increase employment opportunities [85, 91], increase autonomy over nutrition choices [92], support or assess online inclusion [77, 98, 99], increase home and living independence [39], enhance social situations and communication [68, 150], and increase self-determination during independent navigation [79, 88]. The common vision was to increase independence (being able to carry out tasks or responsibilities without relying on others for assistance), interdependence (mutual reliance between individuals or groups), autonomy (having control over your actions and choices), and self-determination (the ability and opportunity to set and pursue your own goals) [36].

Supporting education and therapy is closely related to providing support in accessing opportunities; however, this category is specific to learning or practicing skills. In educational and therapeutic

settings, technology serves as both a supplement and an alternative to traditional instruction, through interactive tools such as games [13, 20, 29] or social robots [4]. They are typically used in a classroom or therapy setting and include professionals in the design and customization process. Technology was used to enable personalized learning and additional scaffolding that helps individuals develop cognitive [13, 29], literacy [54, 70], or dance skills [148]. The use of technology in therapy, such as cognitive [13] or physical rehabilitation [104], creates opportunities for guided practice and feedback that can support progress within and outside of in-person sessions.

Enabling personalization and customization for participants was a common rationale for using technology across various contexts. Personalization refers to the system adapting in real-time to the abilities and needs of each user. Customization allows users, or their supporters, to adjust settings and features according to their personal preferences. For AwDS, this flexibility is essential to ensure that technologies remain accessible, engaging, and effective. For instance, customizable AAC devices can be tailored to accommodate different literacy levels and communication styles [70], while smart home technologies can be adapted to suit a range of living preferences and routines [39]. Similarly, custom route plans with step-by-step images can support independent navigation in community settings [79, 88]. Mobile apps often support personalization by allowing user preferences to evolve over time [92]. Features such as selecting an avatar, choosing rewards for goal completion, and involving preferred team members can help increase motivation [92]. The ability to adjust the content and difficulty level based on individual preferences and abilities also improves outcomes in therapeutic and educational settings [13, 29], ensuring each user's individual needs are met.

Enhancing engagement or motivation was both referenced as a reason to explore technology rather than non-technological options and an outcome for participants. Robots that respond to speech, use gestures, and employ repetition [4], or platforms with gameplay interaction [13, 20, 29] have the power to turn mundane or challenging tasks into enjoyable and rewarding experiences. By integrating interactive elements such as rewards, video encouragement from loved ones, or real-time feedback [92], these technologies encourage users to stay focused and motivated to continue. Within the education and therapy contexts, sustained engagement and motivation are essential for effective learning or skill acquisition [13, 20, 29, 54, 148].

While these categories help organize the key motivations behind employing technology, they are not mutually exclusive. Many of the reviewed publications referenced multiple overlapping goals, for instance, a tool designed to support education also incorporated personalization features or aimed to increase user motivation through gamification. This overlap reflects the multifaceted role technology can play in supporting AwDS across areas of life. Although the categories identified capture the primary reasons described across the 20 studies, they are not intended to be exhaustive. Rather, they provide a framework for understanding the core drivers behind the use of technology in research with AwDS and emphasize the importance of designing flexible, multipurpose tools that can accommodate a range of user preferences and goals. They also highlight gaps that have yet to be explored with AwDS.

5.3 RQ3: How Have Technologies Been Applied to Support AwDS, and in Which Life Domains Are These Applications Most Common?

To answer RQ3, we first examined how technology was applied or researched in the studies and the areas of life it aimed to support. Among the 20 publications, we identified four primary applications of technology: (1) providing general support in an area of living; (2) teaching skills or concepts; (3) examining the accessibility, usability, or adoption of mainstream technologies; and (4) practicing previously learned skills. These technologies were used across six key areas of daily

life: (1) independent living; (2) cognitive, memory, and learning assistance; (3) communication and literacy; (4) independent navigation; (5) workplace assistance; and (6) health and nutrition.

5.3.1 Application of Technology. Across the 20 publications included, we identified four main applications of technology: (1) providing general support in an area of living ($n = 8$); (2) teaching skills or concepts ($n = 7$); (3) examining the accessibility, usability, or adoption of mainstream technologies ($n = 5$); and (4) practicing previously learned skills ($n = 4$).

The first and most common purpose was to *provide general support* in daily life and assistance while performing various tasks. This category is about using technology as an assistant, as opposed to using it as a learning tool, by providing real-time feedback or step-by-step guidance. These technologies assist in navigating daily life by breaking down complex actions, such as making navigational choices [79, 88], managing nutritional choices [92], managing work tasks [85, 91], or living independently in an apartment [39]. Communication technologies (i.e., AAC) also fall into this category [68, 150], as they support AwDS in expressing themselves.

Some technologies were instead used to *teach skills or concepts*, either alongside an expert (e.g., an educator or therapist) or independently. These tools were designed to introduce new content and guide users through initial learning, often using interactive formats to increase engagement. The skills being taught ranged from literacy and language [4, 54, 70], dancing [148], counting [20], or practical daily life skills [39, 92]. These technologies are often interactive, utilizing games, virtual environments, or robotics to engage AwDS, all aiming to introduce and explain new ideas or behaviors.

Technologies were also used for *practicing previously learned skills*. While closely related to learning tools, these technologies were not designed to introduce new content; rather, they help users rehearse, reinforce, or generalize skills over time. In several studies, practice occurred after initial instruction from a teacher or therapist, with participants using the technology independently to revisit tasks and improve performance. These systems included structured cognitive games, interactive simulations, and personalized feedback tools designed to support skill retention and real-world application [13, 29, 89]. Video modeling was also used to guide users through task sequences as a way to reinforce routines [104]. These technologies emphasized repetition and autonomy, enabling users to engage in ongoing practice beyond the learning environment.

Another application researchers employed was the *assessment of accessibility, usability, or adoption* of mainstream technologies. This included evaluations of workplace tools, such as social media, e-mail, scheduling, price comparison, and note-taking [85]. Other researchers investigated the feasibility of speech recognition [68] and authentication methods [99]. Broader digital participation was also explored, such as how AwDS engage with technologies across various aspects of their lives, including home, transportation, entertainment, work, and social interactions [98], and how this usage relates to their overall well-being [77]. Unlike the other categories, these studies did not aim to provide support or instruction; instead, they sought to understand current experiences and barriers to access.

5.3.2 Ability Enhanced through Technology. Building on our understanding of how technology is applied, we identified six key areas where researchers have leveraged technology to assist AwDS: (1) independent living ($n = 5$); (2) cognitive, memory, and learning assistance ($n = 5$); (3) communication and literacy ($n = 5$); (4) independent navigation ($n = 3$); (5) workplace assistance ($n = 2$); and (6) health and nutrition ($n = 2$).

Independent living was the focus of five studies, whether through daily task assistance, skill instruction, or broader lifestyle support. Dalprà et al. [39] piloted a smart home environment, “Casa Satellite,” to explore how home automation could assist AwDS in managing everyday household routines. Similarly, Mechling et al. [104] evaluated the use of video modeling as a method to teach

essential home living skills, specifically examining how individuals performed tasks requiring different levels of motor coordination. Several papers addressed aspects of independent living by considering digital participation and access. Lussier-Desrochers et al. [98] investigated how AwDS engage with digital tools in everyday contexts such as commuting, home life, and communication. Their work emphasized the need to promote digital inclusion to reduce the risk of social exclusion and support autonomy. Ma et al. [99] examined the usability of authentication systems for online services, highlighting the importance of designing secure and accessible interfaces that AwDS can use independently. Jevne et al. [77] explored quality of life from the perspective of AwDS, identifying autonomy, self-determination, and the ability to live independently as essential components of well-being. Together, these studies highlight that independent living is supported not only by specific tools or environments but also by broader digital access and self-directed use of technology.

Cognitive, learning, and memory supports were equally explored in five studies. Alemi and Bahramipour [4] used robot-assisted language learning to improve English vocabulary acquisition and retention, supporting both learning and memory in AwDS. Bargagna et al. [13] developed STELLA DS, a web-based tool designed to strengthen cognitive abilities such as attention and language while offering accessible, at-home brain training. Similarly, Buzzi et al. [29] created a platform that targeted memory, logical reasoning, and orientation through interactive games, enhancing overall cognitive engagement. Villamil Matallana and Paredes-Velasco [148] focused on supporting learning through movement by introducing a dance-based tool that helped develop motor coordination and creativity. Boleracki et al. [20] reinforced numeracy skills by designing a virtual counting game tailored to the user's abilities. Across these studies, technology was used to promote cognitive growth, skill development, and sustained engagement in learning activities.

Communication and literacy skills was also explored in five studies. Alemi and Bahramipour [4] used a robot-assisted learning program to support English vocabulary development, combining repetition, real-life scenarios, and interactive engagement to language skills. Holyfield et al. [70] examined the use of an AAC app feature to support decoding and encoding skills. Gallaher et al. [54] explored the integration of computer-based instruction with tutoring strategies to support phonological awareness, sight word recognition, and phonics skills, showing how technology can be embedded in comprehensive literacy instruction. Holyfield and Drager [68] investigated the feasibility of incorporating natural speech into AAC systems, analyzing intelligibility and consistency to inform the design of more inclusive communication tools. Wilkinson and Bennett [150] focused on the design of AAC displays, demonstrating that layout and visual structure can influence communication efficiency and flexibility. These studies collectively illustrate the role of technology in supporting diverse communication preferences and literacy learning goals for AwDS.

Independent navigation was explored in three studies and is presented as a distinct category from broader independent living due to its recurring emphasis and specific design focus. While navigation is an important component of daily life, these studies explicitly framed it as a discrete challenge requiring targeted technological intervention. Moreover, unlike general household or lifestyle supports, navigation technologies often involve geolocation, real-time decision-making, and mobility planning, skills particularly relevant to adults seeking greater autonomy in the community. For example, Khan et al. [79] examined barriers to independent travel and investigated how smart devices and intelligent technologies might address these challenges. Their work emphasized the importance of understanding real-world travel obstacles and gathering user-informed design requirements to develop more supportive navigation tools. Lachapelle et al. [88] tested applications to help users travel independently between multiple city locations. Their study also highlighted how these tools could support broader decision-making and problem-solving skills related to navigation. While more focused on assessment than intervention, N'Kaoua et al. [89] tested assistive procedures

addressing different aspects of navigational processing. Their findings discussed the importance of targeting route learning and spatial awareness to improve independent travel skills.

Workplace assistance through technology within office-based roles for AwDS was explored in two studies. Kumin et al. [85] examined how usability improvements on touch devices could support tasks such as emailing, scheduling meetings, socializing, and note-taking in a professional setting. Similarly, Lazar et al. [91] observed how experienced technology users with DS engaged with workplace computers, focusing on web searching, e-mail, social networking, and productivity applications. Their study also examined input methods, such as keyboard and mouse use, and included opportunities for participants to demonstrate their preferred tools and workflows.

Health and nutrition supports, focusing on promoting healthier lifestyles and physical engagement, were explored in two studies. Lazar et al. [92] performed a co-design with AwDS and stakeholders to develop an app that teaches healthy food choices and enables users to track their selections over time in pursuit of personal goals. Villamil Matallana and Paredes-Velasco [148] introduced “MyDanceDown,” a smartphone app designed to teach ballet movements, supporting physical activity and creative expression in both classroom settings and at home.

Technology was applied in multiple ways to each life domain, as shown in Figure 6. Within independent living, researchers have typically explored digital inclusion, but also examined general support, teaching, and practice. On the other hand, cognitive, learning, and memory supports were designed as tools for teaching and practice. Researchers focusing on communication and literacy skills were exploring teaching tools, general support options, and digital inclusion. Independent navigation was explored through support and practice. Workplace assistance was explored through digital inclusion and general support. Health and nutrition were emphasized as teaching tools, as well as a source of general support.

5.4 RQ4: What Benefits and Challenges Have Been Reported in the Use of Technologies by AwDS?

5.4.1 Reported Benefits of Technology Use. Across the 20 studies, a range of benefits were reported when AwDS engaged with technology, shown in Table 3. Rather than focusing solely on task success, these findings highlighted how well-designed technologies can support (1) confidence, enjoyment, and engagement ($n = 11$): technologies impacted emotional and motivational outcomes, including pride, interest, and sustained use; (2) independence and digital inclusion ($n = 11$): technologies supported autonomy in daily routines, self-management, and participation in digital life; (3) skill development and learning ($n = 10$): technologies supported measurable growth in areas such as literacy, memory, or navigation; (4) communication through writing, typing, and verbal expression ($n = 9$): technologies supported expressive abilities, such as writing, speaking, or using AAC; and (5) social inclusion and connection ($n = 8$): technologies extended beyond expression to emphasize relationships, interaction, and shared experiences. While we present these as distinct categories, many studies reported benefits that spanned multiple areas. For example, communication tools often supported social interaction, and learning technologies frequently contributed to motivation and enjoyment. We chose to separate these categories to reflect the specific dimensions of benefit emphasized in each study, while acknowledging that technology use is inherently multidimensional and interconnected.

Confidence, Enjoyment, and Engagement. A common thread across studies was the role of technology in boosting motivation, enjoyment, and confidence among AwDS. Participants consistently reported positive emotional responses to technology use, from pride in their achievements to excitement about participation. Educational and communication systems frequently supported both skill development and emotional investment. In Holyfield et al. [70], participants shared that

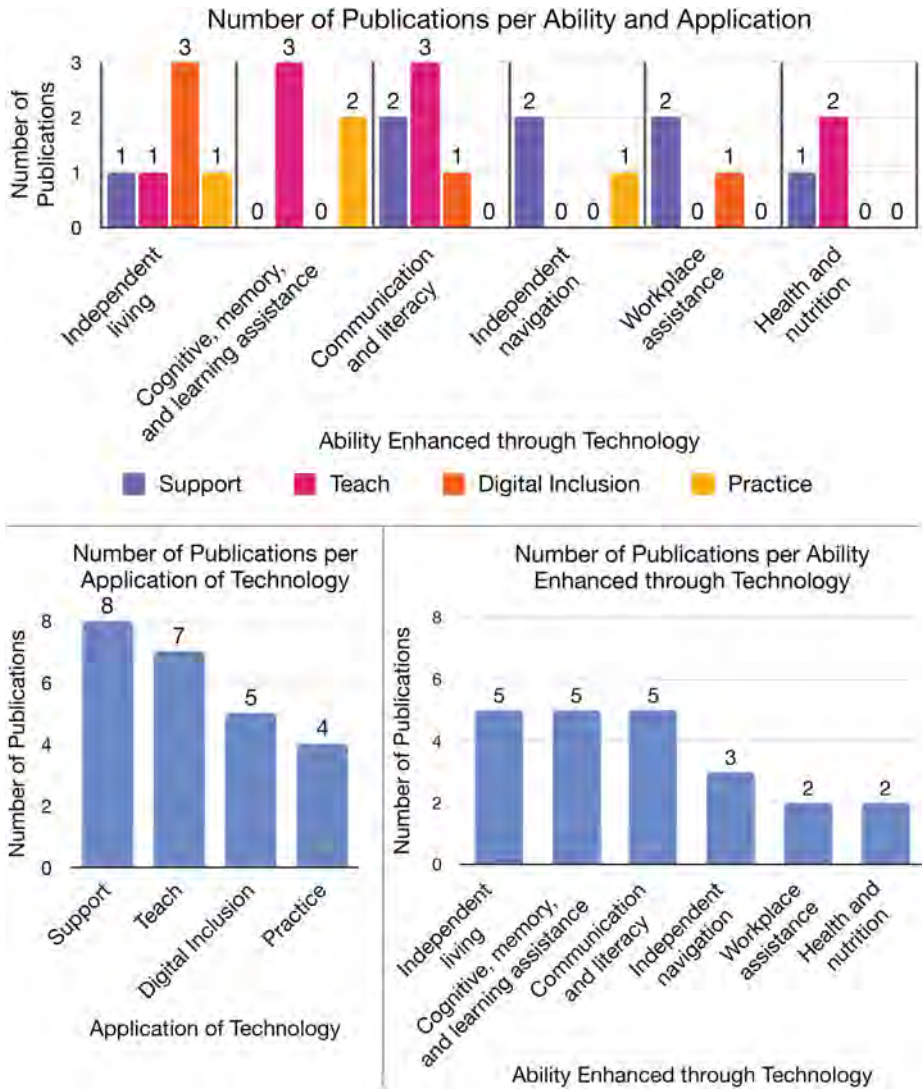


Fig. 6. *Top*: This bar graph shows the number of publications for both the application of technology and the ability enhanced through technology. *Bottom left*: Bar graph illustrating the number of publications by technology application category. Sixteen publications were assigned a single category, while four were assigned two categories. Support and Teach ($n = 2$); Support and Digital inclusion ($n = 2$). *Bottom right*: Bar graph showing the number of publications by supported ability category. Eighteen publications were assigned a single category, while two were assigned two categories. Cognitive, memory, and learning support, and Health and nutrition ($n = 1$); Cognitive, memory, and learning support, and Communication and literacy ($n = 1$).

they enjoyed sounding out words with the computer and felt proud of their reading progress. Similarly, the participant in Gallaher et al. [54] expressed motivation to explore the internet, write drafts, and rehearse identification skills, while remaining engaged across a full year of digital literacy activities. Buzzi et al. [29] observed that users were enthusiastic about a gamified cognitive platform, refusing pauses and responding positively to increasing difficulty levels. Participants also

appreciated attractive visuals and tailored feedback [13, 29, 92]. Motivation was particularly strong when technologies were interactive, personalized, or social in nature. In Alemi and Bahramipour [4], students working with a humanoid robot showed increased classroom participation, attempted to attract the robot's attention, and mimicked its gestures. The repetition, tone of voice, and integration of motion-based activities contributed to both engagement and comfort. Participants also enjoyed group and individual learning with the robot, often interacting more confidently in English. Villamil Matallana and Paredes-Velasco [148] similarly found a link between student motivation and skill development in a dance learning app. Positive emotions were not limited to instructional contexts. In Jevne et al. [77], participants reported eagerness to use smartphones and tablets. In navigation-focused studies, participants and caregivers described excitement, confidence, and positive feelings during technology-supported travel [88]. Caregivers and self-advocates in Lazar et al. [92] responded favorably to motivational elements such as praise, token rewards, and visual stimuli like color-coded cues.

Independence and Digital Inclusion. Across multiple studies, digital technologies were shown to play a valuable role in supporting autonomy and digital inclusion for AwDS. Technology helped participants manage everyday routines, build independent living skills, and increase confidence in navigating digital environments. In a supported living context, participants demonstrated long-term improvements in organizing schedules, preparing meals, and shopping, with one participant describing their smart-enabled apartment as “a home where it is possible to do yourself” [39]. Participants set alarms to wake up, used tablets to organize tasks, and relied on phones to stay in contact with caregivers or request help when needed [77, 79]. Wearable and discreet technologies were also identified as preferential for preserving safety and independence during solo travel [79]. Technologies further supported digital participation and leisure. AwDS frequently engaged in informal activities like watching videos, reading online, or exploring interests on their devices [54, 77]. Many reported using phones and tablets for entertainment and social connection, often through platforms like Facebook, Snapchat, and Messenger [77, 98]. These tools allowed users to stay up to date, send messages, and plan events, mirroring everyday digital routines of the general population. Others expressed a desire to expand their digital skills, from searching for recipes to managing their own online activity [98]. Participants benefited from tools that offered search boxes, autosuggestions, consistent visual cues, visual feedback, and scheduling features [77, 85, 88]. Participants in Lachapelle et al. [88] completed solo travel using step-by-step visual and audio prompts, with caregivers reporting increased confidence and problem-solving ability. Digital fluency was also observed in structured and professional settings. Expert participants in Lazar et al. [91] used multiple software systems across devices, managing documents, calendars, spreadsheets, presentations, and databases with minimal support. They also showed strong use of formatting tools, spell check, and calendar coordination, often across shared family platforms. In studies on authentication, AwDS successfully used traditional password systems and, with training, were able to increase both speed and accuracy of entry [99]. Participants performed comparably to or even outperformed neurotypical peers in error rates, demonstrating that password entry does not have to be a barrier to access.

Skill Development and Learning. A benefit of technology use among AwDS was the development of new skills and reinforcement of existing ones. Across studies, digital tools were shown to support cognitive, motor, and daily living skills, both through structured instruction and embedded use in everyday routines. Several studies reported measurable gains in educational or training contexts. In a robot-assisted language learning study, Alemi and Bahramipour [4] found that participants significantly outperformed a control group in post-test performance, particularly when learning English verbs. Similarly, a literacy AAC app in Holyfield et al. [70] helped participants improve word

decoding skills, with all users reporting reading gains. Participants in Mechling et al. [104] completed more fine motor steps independently when guided by video modeling, suggesting that visual learning strategies can improve practical task execution. Daily living skills were also strengthened through the use of technology. In a longitudinal smart home study, Dalprà et al. [39] found that participants regained or built upon prior knowledge, emphasizing the potential for technology to reinforce skill maintenance. Skills related to navigation and spatial understanding were supported through a multimodal design approach. In N’Kaoua et al. [89], landmark-based prompts improved performance in wayfinding and route-learning tasks. While the aids did not support deeper configurational understanding, they did enhance immediate task success. In the creative domain, Villamil Matallana and Paredes-Velasco [148] observed a positive trend in dance skill development, although the improvement was not statistically significant, motivation and practice were closely linked to skill acquisition. Skill development also extended to digital competence and security. Ma et al. [99] demonstrated that AwDS learned and used multiple forms of password-based authentication. Although graphical passwords initially presented challenges, users improved over time, supporting the case for learning-oriented security features. Participants who practiced new systems gained both speed and reliability in their interactions. Together, these findings suggest that digital technologies, when thoughtfully designed and embedded into everyday contexts, can promote learning for AwDS. The benefits extended across formal and informal settings, from educational and home environments to social and professional life.

Communication through Writing, Typing, and Verbal Expression. Communication technologies enabled AwDS to express themselves through multiple modalities, including writing, typing, and speaking. These tools provided opportunities to utilize expressive skills for personal, educational, and occupational tasks. For example, touchscreen and external keyboards were commonly used to compose messages, manage content, and respond to others [85, 91]. Participants demonstrated typical typing speeds on a touchscreen when compared to an external keyboard, preferred replying to messages over initiating them [85], and skillfully used formatting tools, spell check, and document editing features [91]. Communication also played a role in organizing daily life as participants used calendars, reminders, and coordinated messages with family members to manage schedules [77, 91]. In structured contexts, communication tools also support skill development. Alemi and Bahramipour [4] reported increased motivation to communicate in English through robot-assisted learning, and Gallaher et al. [54] observed a strong desire to use digital tools for writing and information searching. Speech-based systems also showed promise, with studies highlighting consistent speech production [68] and improved communicative acts when using optimized AAC displays [150]. E-mail, texting, and video calling were widely used across devices such as smartphones and tablets [77, 98]. Participants used platforms like Facebook, Snapchat, and Messenger, with several noting that messaging felt easier than calling [77].

Social Inclusion and Connection. Digital technologies played a meaningful role in facilitating social interaction, helping AwDS maintain relationships and share experiences. Technologies facilitated the emotional and relational aspects of staying connected. Communication tools, such as text messaging, video calls, and social media, enable participants to stay in touch with family, friends, and broader networks. In Jevne et al. [77], participants used platforms such as Facebook, Messenger, Snapchat, and Instagram on a daily basis, describing them as important for staying updated, making plans, and keeping in touch with loved ones. These tools also extended to leisure and casual interaction, as phones and tablets were used for amusement and everyday check-ins [77, 85]. In Lazar et al. [91], participants used shared calendars and e-mail coordination to plan events with family members and coworkers. Multiple e-mail accounts and cross-platform communication reflected strong digital engagement and adaptability. Similarly, Lussier-Desrochers

Table 4. Identified Challenges of Technology Use across Studies

Citation	Interface Complexity	Context-Aware Interaction	Uneven Impact	Environmental and Sensory Constraints
Kumin et al. [85]	x	-	x	-
Alemi and Bahramipour [4]	x	x	-	-
Lazar et al. [92]	-	x	-	-
Bargagna et al. [13]	x	-	-	-
Lussier-Desrochers et al. [98]	x	x	-	x
Holyfield et al. [70]	x	-	x	-
Mechling et al. [104]	x	x	-	-
Wilkinson and Bennett [150]	x	-	-	-
Holyfield and Drager [68]	x	-	x	-
Ma et al. [99]	x	x	x	-
Buzzi et al. [29]	x	-	-	-
Dalprà et al. [39]	-	-	x	x
Gallagher et al. [54]	-	x	x	-
Villamil Matallana and Paredes-Velasco [148]	-	x	x	-
Jevne et al. [77]	-	x	-	x
Khan et al. [79]	x	-	-	x
Lazar et al. [91]	-	x	-	-
Lachapelle et al. [88]	-	-	x	x
N’Kaoua et al. [89]	-	-	x	-
Boleracki et al. [20]	-	-	-	-
Total	11	9	9	5

“x” = challenge documented; “-” = not documented.

et al. [98] found that participants sent messages and made video calls through tools like Skype and FaceTime, with some expressing interest in learning to use additional platforms such as Facebook for more autonomous social interaction. Technologies also supported interaction in face-to-face and assisted social contexts. Lachapelle et al. [88] described a parent’s positive reaction when their son approached a stranger in a subway station for help. In AAC systems, Wilkinson and Bennett [150] found that optimized display designs increased both the frequency and variety of social communicative acts, including responses to others and comments directed toward communication partners. Displays that incorporated spatial organization and were integrated prompted significantly higher interaction rates than standard setups, demonstrating how inclusive design can enhance interpersonal connections. Lazar et al. [92] found that participants responded strongly to social motivators, such as verbal praise and video messages from respected individuals. These forms of encouragement, particularly when tailored to individual preferences, enhanced user engagement and promoted social-emotional connection between users and those around them.

5.4.2 Reported Challenges of Technology Use. While technology offers many potential benefits for AwDS, several studies also reported challenges related to design, usability, and contextual fit, shown in Table 4. These challenges were not framed as limitations of the users themselves, but rather highlighted where technology systems failed to meet the needs, preferences, or lived experiences of AwDS. The reported issues were grouped into four categories: (1) interface and interaction complexity ($n = 11$): difficulties with layouts, buttons, responsiveness, or input methods that disrupted usability; (2) contextual misalignment in technology use ($n = 9$): mismatches between technology assumptions and users’ real-world preferences, routines, or support systems; (3) limited generalization and uneven impact ($n = 9$): variability in effectiveness across individuals and a lack of

long-term or transferable outcomes; and (4) environmental and sensory constraints ($n = 5$): barriers include noise, screen glare, connectivity, or routine disruptions that interfere with consistent use.

Interface and Interaction Complexity. Interface complexity and interaction design were common challenges across multiple studies. Small icons, hard-to-reach buttons, authentication processes, and slow or unresponsive features often interrupted interaction flow. Kumin et al. [85] reported that some participants had difficulty locating small icons on touchscreen devices, often scrolling multiple times before finding a specific function. Touch interfaces also led to accidental taps, and some participants required significantly more time to complete tasks due to navigation issues with the interface. Lussier-Desrochers et al. [98] noted that some participants struggled with using search engines, entering keywords, and navigating websites. Holyfield et al. [70] noted that a decoding feature in a literacy app took several seconds to respond, which disrupted the pacing of reading activities. Bargagna et al. [13] reported that their technology demo instructions were not clear and needed to be replaced with more simplified, shorter visual video guides. Holyfield and Drager [68] reported that speech recognition systems struggled to interpret participants' speech, with recognition rates as low as 1.7%, severely limiting the usefulness of these tools for AAC integration. Wilkinson and Bennett [150] similarly noted that non-optimized AAC displays (i.e., grid format, color cues, and non-integrated into other technologies) led to reduced communication and social interaction. Mechling et al. [104] and Buzzi et al. [29] observed that generalized instructional videos or drag-and-drop activities can slow down interaction and do not always support effective learning.

Contextual Misalignment in Technology Use. Many studies reported that the underlying assumptions embedded in technology design, about how users think, behave, or engage socially, did not always align with the real-world expectations, routines, or preferences of AwDS. Some studies have revealed that design strategies, such as peer competition, traditional web navigation, online demonstrations, the timing of authentication steps, and assumptions about caregiver roles, do not always align with AwDS expectations. Lazar et al. [92] found that peer-based competition within apps reduced interest, and that excessive involvement from caregivers was not desired. In Villamil Matallana and Paredes-Velasco [148], caregivers had difficulty using the educational platform, which likely shaped their perception of its usability and caused inconvenience to the AwDS. In Jevne et al. [77], some participants focused on posting only pleasant messages on social media due to prior exposure to inappropriate or unpleasant content. In a robot-assisted learning study, Alemi and Bahramipour [4] found that learners were disappointed when the robot could not respond to spontaneous questions and, in some cases, began treating the robot more as entertainment than as a teaching tool. Similarly, Ma et al. [99] found that mnemonic or graphical password systems were seen as more cognitively and visually demanding compared to traditional passwords. They also found that some participants required slightly more time to effectively use authentication methods, especially with graphical passwords, and that more errors occurred due to capitalization mistakes than typos. Lazar et al. [91] found that expert users sometimes deleted emails immediately after replying, with some even removing the sent copy. One user needed reminders to read a message fully before deleting it, suggesting that default inbox workflows did not always align with users' habits.

Limited Generalization and Uneven Impact. Many systems demonstrated inconsistent impact across participants or failed to support the transfer of meaningful learning beyond specific tasks. Studies reported considerable variation in user performance and outcomes, highlighting that some users benefited while others did not. These inconsistencies suggest that technologies were not always adaptable to diverse learning styles, skill levels, or real-world contexts, limiting their broader effectiveness. For example, Kumin et al. [85] observed significant differences in the time it took

participants to complete basic tasks, with some taking three times longer than others. Variability in attention to detail, accidental touch inputs, and inconsistent navigation strategies led to frustration for some users. Similarly, Gallaher et al. [54] described a participant who relied on random guessing to progress through a literacy game, undermining the intended educational value. Despite repeated use, there was little evidence that this technology-supported long-term skill transfer or improved phonological awareness. In some cases, technology failed to support growth over time. Dalprà et al. [39] reported a significant regression in daily living skills after a summer break, indicating that the benefits of tech-supported routines may not persist without continuous engagement. Likewise, in a study on dance-based learning, Villamil Matallana and Paredes-Velasco [148] found no statistically significant improvement in skills across the group, and noted that caregiver challenges using the tool may have affected learner outcomes. These findings emphasize that systems must account for sustained engagement and flexible reinforcement strategies to ensure benefits are retained and applied across contexts. Ma et al. [99] highlighted that while users with DS learned authentication systems, many defaulted to weaker or insecure passwords (e.g., their own names) or struggled with visual and cognitive complexity. Even after training, password creation and entry remained a barrier for some users, demonstrating that success may depend on both design features and sustained practice.

Environmental and Sensory Constraints. Environmental and sensory constraints, including issues related to difficulty hearing audio, challenges with touchscreen responsiveness, glare from sunlight, time away from structured routines, and concerns about personal safety, were another consistent barrier. Lussier-Desrochers et al. [98] reported that participants often had trouble hearing device audio and adjusted the volume to maximum, while others struggled with touchscreen responsiveness, either pressing too hard or not receiving tactile feedback. Technical issues, such as weak Wi-Fi, forgotten passwords, software updates, or viruses, were also common and occasionally required family support to resolve. Dalprà et al. [39] observed a notable regression in daily living skills after a summer break, suggesting that long breaks in technology-supported routines can affect retention. Lachapelle et al. [88] described how public noise, glare from sunlight, and screen timeout features disrupted navigation tasks, while Khan et al. [79] found that maps with too much information overwhelmed users. Participants in both studies had to pause or move aside in public spaces to use devices, and caregivers expressed concern about carrying expensive or attention-drawing smartphones during independent travel.

5.5 Design Recommendations

Across the 20 reviewed studies, a range of design suggestions emerged to guide the development of inclusive technologies for AwDS. These recommendations were identified from the benefits and challenges shown in Tables 5 and 6, technology design choices, and recommendations within the 20 publications. The literature revealed five interrelated design themes: (1) reduce interaction complexity through clear visuals and simplified choices ($n = 10$); (2) support engagement through responsive feedback and user-controlled pacing ($n = 10$); (3) embed design into everyday routines and environmental contexts ($n = 10$); (4) design for individualization, adaptability, and growth over time ($n = 8$); and (5) provide flexible, user-friendly pathways to secure system access ($n = 2$).

5.5.1 Reduce Interaction Complexity through Clear Visuals and Simplified Choices. Visually clear and simplified interfaces were found to support user navigation, task initiation, and confidence across multiple contexts. Kumin et al. [85] emphasized the value of large, clear icons and cautioned against using small touch targets and pull-down menus, which can slow navigation. Lazar et al. [92] suggested screens should feature limited options, such as no more than nine items per screen, and features like progress bars and traffic light color-coding to reinforce choices. Visually intuitive

Table 5. Mapping of Design Recommendations to Reported Benefits

Design Recommendation	Confidence and Engagement	Independence and Digital Inclusion	Skill Development	Communication	Social Inclusion and Connection
Reduce interaction complexity through clear visuals and simplified choices	x	x	-	x	-
Support engagement through responsive feedback and user-controlled pacing	x	-	x	-	-
Design for individualization, adaptability, and growth over time	-	x	x	-	x
Provide flexible, user-friendly pathways to secure system access	-	x	x	-	-
Embed design into everyday routines and environmental contexts	-	x	x	x	x

An “x” indicates where a recommendation directly supports that benefit area.

Table 6. Mapping of Design Recommendations to Reported Challenges

Design Recommendation	Interface Complexity	Context-Aware Interaction	Limited Generalization	Environmental Constraints
Reduce interaction complexity through clear visuals and simplified choices	x	-	-	x
Support engagement through responsive feedback and user-controlled pacing	x	-	x	-
Design for individualization, adaptability, and growth over time	-	x	x	-
Provide flexible, user-friendly pathways to secure system access	x	-	x	-
Embed design into everyday routines and environmental contexts	-	x	x	x

An “x” indicates where a recommendation directly responds to that challenge area.

layouts and icon-driven navigation were especially important for browsing and communication tasks [54, 150]. Similarly, travel support apps in Khan et al. [79] and visual cue systems in Lachapelle et al. [88] demonstrated how pop-up messages, icons, and simplified maps helped AwDS progress through real-world routines with greater ease. The use of search features with auto-suggest, as noted in Kumin et al. [85], further improved speed and accuracy in locating information, particularly when paired with familiar symbols and minimal visual clutter. Gallaher et al. [54] and Lussier-Desrochers et al. [98] both emphasized how icons and visual cues were relied on for navigating web interfaces, while excessive layers or small links led to confusion. Similarly, Khan et al. [79] reported that overly detailed maps created difficulty during navigation, reinforcing the need for simplified, linear visual layouts. However, it is important to note that not all participants required simplified interfaces to succeed. In Lazar et al. [91], expert users with DS were able to navigate standard productivity software, prompting the authors to note that interface modifications were not needed for most

applications in that context. Although they caution that this finding should not be overgeneralized, it raises an important nuance: the need for simplification may depend on users' prior experience and digital fluency, rather than being a universal requirement.

5.5.2 Support Engagement through Responsive Feedback and User-Controlled Pacing. Multisensory feedback and flexible pacing supported learning, motivation, and task completion. In literacy apps, dynamic movement, audio output, and highlighted visuals helped users follow along with decoding activities and supported engagement [70]. Robot-assisted learning tools provided positive verbal reinforcement, movement-based prompts, and emotional tone to sustain interest [4]. AwDS in Bargagna et al. [13] responded well to feedback that appeared on the screen and was simultaneously voiced aloud, and those in Villamil Matallana and Paredes-Velasco [148] improved performance through teacher-created videos that provided personalized corrections. Some systems also used pre-recorded voice prompts, rather than synthesized speech, to increase user comfort during interaction [13]. Many systems allowed users to complete tasks at their own pace, pause and resume freely, or repeat steps as needed, features shown to improve comfort and independence in learning scenarios [29, 104, 148]. These systems avoided timers or rigid progressions, instead allowing repetition and practice within a safe and responsive structure. However, two studies point to important considerations. In Mechling et al. [104], video modeling was less effective than expected, possibly impeding performance compared to other formats. This suggests that while pacing and feedback are generally supportive, their implementation must be carefully designed and tested for effectiveness. Likewise, in Gallaher et al. [54], a participant used a strategy of random guessing to trigger positive feedback in a literacy game, revealing a risk that poorly designed feedback loops can unintentionally reinforce ineffective learning strategies. These examples underscore the need for engagement systems to promote genuine progress and understanding, not just user interaction.

5.5.3 Embed Design into Everyday Routines and Environmental Contexts. Context-aware design for everyday use addressed the need for systems to function reliably in public and private settings. Travel tools were particularly effective when they included features like GPS tracking, caregiver alerts, step-by-step progress displays, and linear maps with reminders at key points [79, 88]. AwDS benefited from visual photos of landmarks, help buttons, and voice guidance, all of which increased independence while keeping caregivers informed. In noisy environments or outdoor settings, AwDS sometimes had difficulty hearing prompts or seeing the screen clearly, indicating a need for high-contrast visuals, customizable audio, and reduced reliance on ambient sound [88]. Studies have also shown the importance of designing for technical constraints, such as lost Wi-Fi, screen timeouts, or device maintenance issues [98]. AwDS often relied on caregiver support to resolve these issues or complete setup tasks, reinforcing the need for embedded help systems, low-maintenance interfaces, and shared understanding between users and supporters [91, 148]. AwDS in Jevne et al. [77] also described using phones and tablets to support communication and routines, suggesting that technologies must integrate smoothly into everyday life to sustain use over time. However, two papers highlighted limitations to this recommendation. In Villamil Matallana and Paredes-Velasco [148], caregivers faced difficulties using the system, which made it harder for students to engage, suggesting that embedding systems into everyday contexts also requires attention to caregiver usability. Likewise, Lachapelle et al. [88] reported that environmental constraints, such as glare, noise, and screen inactivity, disrupted use during travel. These findings reinforce that while integration into daily routines is important, successful implementation depends on supporting the surrounding conditions and all users involved.

5.5.4 Design for Individualization, Adaptability, and Growth over Time. Personalization and progression over time were essential for making systems meaningful and sustainable. Several tools

offered customizable features such as user-selected avatars, input methods (text or voice), dietary preferences, reward systems, and color schemes [29, 92]. Platforms like those in Buzzi et al. [29] and Bargagna et al. [13] allowed levels of difficulty to be adapted dynamically based on user responses, supporting progression without overwhelming the user. Buzzi et al. [29] further demonstrated the importance of personalization by implementing errorless learning and allowing both text and voice input depending on user preference. These features were paired with customizable avatars, motivational systems, and home/school continuity, reinforcing the importance of adaptive design frameworks that reflect real-world learning paths. In travel and health apps, AwDS or their caregivers could personalize content, upload images of landmarks, or adjust routines to reflect individual preferences and safety needs [79, 88]. Support for personalization was especially helpful during onboarding, as AwDS often relied on caregivers or professionals to configure technology before transitioning to independent use [92, 148]. Setup wizards, tutorial videos, and co-configured features provided initial scaffolding while enabling eventual independence. Across the reviewed studies, no direct contradictions emerged to challenge the value of personalization or adaptability. While not all technologies implemented adaptive features, no quotes suggested that customization hindered use. Instead, the findings consistently affirmed the importance of tailoring systems to meet user needs and support their growth over time.

5.5.5 Provide Flexible, User-Friendly Pathways to Secure System Access. Secure and accessible system access was another important design feature, particularly for tools that required authentication or personalization. Ma et al. [99] found that AwDS could successfully use mnemonic and alphanumeric passwords, but required slightly longer timeout periods and clearer visual feedback, such as seeing typed characters briefly or receiving Caps Lock alerts. Passwords incorporating familiar names were common, and systems needed to help guide users toward more secure formats without increasing cognitive demands. They also explored the use of graphical passwords, which initially slowed performance but showed potential with practice. These findings suggest that password systems should prioritize cognitive and visual simplicity over efficiency. Lazar et al. [92] further recommended alternatives like Touch ID or third-party logins (e.g., Facebook or Gmail) to reduce barriers. Together, these findings reflect a broader trend in which AwDS values ease of use, clarity, and support in authentication, highlighting the importance of offering flexible login options tailored to diverse user preferences. This design recommendation was not frequently discussed, but it addresses a critical usability barrier and provides important guidance for systems that require secure access.

These five design recommendations demonstrate how technologies can be designed to respond to the abilities, preferences, and routines of AwDS. While the reviewed studies generally converged on the importance of clarity, personalization, and contextual fit, a few inconsistencies emerged in the design recommendations. These differences may reflect variation in participants' prior age-related experience with technology, the presence of support networks, or the specific demands of each task. Similarly, while personalization was widely recommended, the extent and form of adaptation varied. Some studies emphasized caregiver-led customization [148], while others prioritized user-controlled features [29]. These divergences suggest that technology design for AwDS must accommodate a range of user needs and support strategies, rather than relying on uniform approaches.

6 Discussion

This scoping literature review synthesized findings from 20 publications to explore the role of technology with AwDS. Through the analysis, we identified trends in research methods; technologies that have been explored; areas of living that the technologies were designed to support; and the associated benefits and challenges faced by AwDS when using these tools. As recommended by

PRISMA-ScR guidelines, we did not assess risk of bias, which may limit the strength of some conclusions. In this discussion, we reflect on these findings and highlight their implications.

RQ1. We found that researchers employed a range of methodological approaches to study technologies with AwDS. The categories reflect an interdisciplinary and evolving research space, where methods range from controlled studies to co-designed development. Experimental studies were the most commonly used, often structured to assess the performance outcomes of specific interventions [70, 104, 150]. Usability studies emphasized real-world practicality and user experience, frequently testing the functionality of systems by examining factors like task performance, user preferences, and interface design [13, 29, 85, 88, 99]. Exploratory studies focused on feasibility and participant experience, generating early insights into emerging tools, user needs, and underexplored contexts [68, 77]. Participatory design centered user collaboration throughout the design process, producing tools shaped by participant preferences [79, 92]. Finally, observational and ethnographic studies offered rich, contextual understandings of how technologies are used in everyday life [39, 91]. All studies included AwDS, but not all of them focused their data collection on AwDS; instead, they focused on the opinions of stakeholders and experts. We encourage future researchers to highlight their participants' perspectives and respectfully include them in research processes. We also encourage researchers to include AwDS in the consent process, even if some individuals require consent through a legal authority. We noticed a few observational and ethnographic studies, which is surprising given their flexibility and potential to capture natural interactions, especially when participants prefer to demonstrate rather than describe their use of technology. However, within studies using qualitative or mixed-methods approaches, observation was a common data collection element. We also observed a lack of participatory design methods, which may be due to the multidisciplinary contributions. Participatory methods are highly recommended within accessibility research [124, 139]. They are also scrutinized for lacking political clarity and struggling to scale to broader societal challenges [11]. Participatory methods, such as those conducted with focus groups by Lazar et al. [92], may be particularly useful when working with the DS community. However, researchers must consider the time and effort required from participants when using participatory methods. Additionally, expanding the use of ethnographic, longitudinal, or diary-based methods could further diversify the field and promote more authentic inclusion of AwDS in technology research.

Across the 20 studies, researchers reported multiple strategies to ensure inclusive study design. Commonly, stakeholders were involved in some capacity (e.g., caregivers, educators, therapists) and were integrated into the study process to provide feedback, support, or interpretation [4, 68, 148]. Study personalization provided flexible interview formats and tailored interfaces to better align with individual interests and abilities [77, 92, 150]. Several studies adapted task instructions, offered scaffolding, or used multimodal guidance to improve comprehension and task engagement [29, 68, 85]. Creating comfortable and familiar environments, such as conducting research in homes, classrooms, or participant-preferred locations, was also a notable approach to reducing anxiety and increasing participation [4, 77]. Visual aids, such as pictorial flashcards, Likert scales, or storyboarding, were used to support verbal study instructions [68, 92]. However, their use was less prevalent than expected, given the substantial literature emphasizing the value of visual supports for individuals with DS. While visual elements were frequently discussed, they were more often integrated into the technologies being evaluated rather than the research protocols themselves. Warm-ups or training tasks were used to build participant confidence before beginning the main activities [85, 89]. As this field continues to expand, reporting and reflecting on inclusive study design will be critical to building ethical, empowering, and relevant technologies alongside the AwDS community. Despite these inclusive practices, no studies reported including individuals who

used AAC as their primary form of communication, highlighting a need for broader representation in research participation. Additionally, other inclusive strategies, such as physical assistance, are reported in studies involving individuals with IDD [7, 62, 132], but were not observed in the AwDS literature. Similarly, verbal praise and reward-based motivation have been documented in studies involving developmentally diverse children, including those with DS [22, 69]. These strategies were not highlighted in studies focusing on AwDS. Five studies did not report any inclusive strategies, which may reflect that some adults do not prefer or require specialized support. We recommend that researchers proactively prepare to offer a range of inclusive strategies, allowing participants to choose options based on their individual preferences and needs.

RQ2. The technologies studied across the 20 reviewed publications were diverse. Computers were frequently used as platforms for educational content [13, 54], workplace tools [91], and simulated environments for practicing independent living skills [20, 89]. Video modeling was applied to reinforce skill acquisition through structured repetition [104]. These systems were often selected for their flexibility and ability to support structured, screen-based learning tasks. A small number of studies incorporated AAC technologies, though these were all computer-based and often served as support to the study's main focus. These systems included symbol displays embedded in storybooks [150], literacy-focused decoding supports [70], and early experiments with speech recognition integration [68]. Commonly explored tools also include smartphone and tablet applications [79, 88, 92, 148], often chosen for their portability, familiarity, and suitability for use across everyday contexts. These mobile platforms supported a wide range of activities, including navigation, scheduling, nutrition tracking, and communication. Smart home systems were introduced to support activities of daily living through interactive environmental control [39]. One study also evaluated a social robot as a teaching assistant for language instruction [4].

Despite the range of tools explored, several promising technologies from prior accessibility research were notably absent. No studies investigated wearable technologies, TUI, VR/AR, or smart speakers. Additionally, technologies supporting financial literacy or mental health, which are increasingly relevant for adult autonomy, were not represented. These gaps point to valuable directions for future exploration, especially as prior IDD-focused reviews have shown that tools like VR/AR and internet-based platforms can increase confidence, support communication, and promote social engagement [55, 108]. In contrast to reviews focused on children with DS, which tend to emphasize games and cognitive training tools [19, 131], the studies in this review placed less emphasis on game-based learning, potentially signaling a shift away from play-centered design toward tools aimed at supporting everyday functioning and autonomy in adulthood.

Researchers identified four main motivations for exploring these technologies. The most common motivation researchers expressed for exploring digital technologies was to support access to real-world opportunities, such as employment [85, 91], social inclusion [98, 99], independent living [92], and navigation [79, 88]. Education and therapy supports, focused primarily on cognitive, motor, communication, and literacy [13, 29, 70, 148]. Technology was developed to individualize learning experiences and facilitate feedback from educators or therapists. Customization and personalization were frequently cited as reasons for studying technology. Tools incorporated features like custom avatars or rewards [92], and tailored route plans or learning modules to suit individual needs [13, 79]. Technology was also selected with the goal of increasing user engagement and motivation. Gamified platforms, social robots, and interactive feedback systems transformed learning and daily routines into more enjoyable experiences [4, 20, 92]. These features were particularly emphasized in educational contexts, where sustained attention and motivation are critical for learning outcomes [54, 148]. While these four motivations reflect the primary goals described across the reviewed

studies, they also highlight the field's current focus on direct task support, education, and communication. As new technologies emerge, other motivations may become more prominent, such as supporting emotional expression, fostering creativity, enabling self-advocacy, or strengthening social identity. For instance, immersive technologies, more commonly seen in studies with broader IDD populations [7, 62, 105, 136], or creative platforms like video blogs and augmented storytelling apps could shift motivations toward exploration, play, or self-expression, rather than task completion alone. Immersive systems were not represented in the AwDS-focused studies. It remains unclear whether this absence reflects the small sample of studies or a difference in participant preferences and research priorities. The AwDS literature emphasized more real-world, contextually grounded interventions, suggesting a strong orientation toward practical and functional support. Similarly, future research on wearable or sensor-based tools may open new possibilities for real-time emotional regulation, health awareness, and other person-centered goals. Broadening the range of technologies studied could help uncover new motivations that more fully reflect the aspirations and lived experiences of AwDS.

RQ3. The reviewed technologies were applied in four distinct ways, each serving a different function in supporting AwDS. The most common application was to provide general support, where technology acted as an assistant in everyday contexts, offering real-time guidance, structured routines, or communication support. These tools helped facilitate navigation, independent living, and workplace participation through step-by-step task management [39, 79, 85, 88]. In contrast, teaching technologies were designed to introduce new skills or concepts, often in collaboration with educators or therapists, using interactive formats such as games, robots, or virtual environments [4, 54, 148]. A related but distinct category involved technologies that supported ongoing practice, emphasizing repetition and feedback to reinforce skills after initial instruction. These tools promoted memory, routine development, and independent task repetition over time [29, 104]. Finally, some studies have focused on evaluating technology use, assessing the accessibility, usability, or adoption of mainstream tools, such as workplace software, authentication systems, or information and communication technologies [77, 85, 98, 99]. Unlike the other categories, these studies were less intervention-focused and aimed to understand real-world usage patterns and inclusion gaps. Using technology to teach specific skills was a prominent focus reflected by the influence of special education, healthcare, and communication sciences departments, which comprised a large portion of contributing authors. It also signals a continued opportunity for HCI and accessibility researchers to further engage with issues of digital inclusion beyond clinical or therapeutic contexts.

Across these four applications, technologies were deployed to support AwDS in six life domains. Independent living was the primary focus, with tools designed to promote autonomy in home, public, and online settings. For example, smart home technologies or video modeling assisted with routine tasks [39, 104], while authentication and web navigation studies explored how to make digital tools more secure and accessible [99]. Broader lifestyle integration was also emphasized through studies of digital inclusion and user experiences [77, 98]. Navigation tools, such as mobile apps and virtual environments, helped users plan routes and make travel decisions [79, 88, 89], often incorporating user-informed design to better align with real-world needs. In workplace settings, technologies such as tools for communication, scheduling, and documentation have supported AwDS in office environments [85, 91]. Health and nutrition-tracking applications enabled healthy eating habits [92], and a dance-based movement app promoted physical activity and creative expression [148]. Cognitive, learning, and memory supports were often developed on platforms designed to build language, numeracy, attention, or reasoning skills [4, 13, 20, 29, 148]. Often blending structured instruction with personalization and user engagement. Similarly, communication and literacy development was another area supported through AAC systems, reading tools, and language-based

games [4, 54, 68, 70, 150]. Few tools directly addressed social skill development, possibly due to the perception that AwDS often have strong social motivation and skills [52]. This differs from prior IDD research, where supporting social interaction is often a design goal [90, 108, 154]. While these themes capture the core motivations within the reviewed studies, they are not exhaustive. Technologies can also support self-expression, creativity, financial management, health monitoring, and access to news or entertainment, areas that remain underexplored in the context of AwDS and may offer promising directions for future research. Notably absent from this body of work were technologies supporting relationships, sexuality, or marriage, topics that emerged in reviews on adults with IDD, such as Glencross et al. [55], but were not reflected in this dataset. This highlights an ongoing need to ensure that research priorities reflect the full scope of adult life experiences and identities. Continued investment in inclusive, context-aware design will be essential to ensure these technologies remain relevant, empowering, and widely accessible.

RQ4. Across the reviewed studies, digital technologies provided a wide range of benefits beyond task performance alone. Technologies that supported autonomy and digital inclusion enabled participants to independently manage routines, build life skills, and engage in everyday digital practices, from setting alarms and organizing grocery lists to coordinating calendars and navigating solo travel [39, 77, 79]. Many participants demonstrated strong digital competence when supported by consistent interfaces and training, managing documents, spreadsheets, e-mail accounts, and authentication systems across devices [85, 91, 99]. Communication tools facilitated expression across written, verbal, and visual modalities, with several studies reporting high engagement through optimized AAC systems, speech technologies, and message-based platforms [68, 85, 150]. Social inclusion was another consistent theme. Messaging, social media, and collaborative digital tools enabled users to stay connected with caregivers, family, and friends, while also participating in broader online communities [77, 98]. Technology also fostered confidence, enjoyment, and motivation. Many users described pride in their progress, motivation to engage with content, and enjoyment from interaction, especially when tools incorporated meaningful feedback, appealing visuals, or personalized supports [4, 29, 70]. In education and daily living contexts, technologies helped reinforce skill development, from language and reading to navigation, password use, and dance practice, often through repetition, scaffolding, and multimodal feedback [4, 89, 148].

Despite these benefits, researchers also identified challenges that limited usability and adoption. Interface complexity, such as small icons, slow feedback, or confusing navigation, was a frequent issue, disrupting task flow and increasing reliance on support [70, 85]. Technologies involving speech input failed to accommodate the speech patterns of AwDS, leading to frustration or reduced effectiveness [4, 68]. There was a contextual misalignment of users' needs or preferences in technology use, including built-in peer competition, complex password systems, or web-based tutorials, which led to reduced motivation or autonomy [13, 92, 99]. Despite the learning growth highlighted through the studies, there was limited generalization and an uneven impact across users. Even when technologies were designed with accessibility in mind, not all users benefited equally, and some experienced minimal gains or even regression without consistent reinforcement [39, 54, 148]. Several systems supported task completion but did not lead to deeper learning, sustained independence, or skill transfer across contexts [89, 91, 99]. Finally, environmental factors, like screen glare, background noise, or security difficulties, also interfered with use, particularly in public or unstructured settings [39, 88]. These challenges highlight the importance of flexible systems that accommodate individual variability and support long-term growth.

Together, the reported benefits and challenges reveal a web of interconnected themes, with social connection emerging as a particularly strong throughline, even in studies not explicitly focused on social interaction. This was evident in the close relationship between the benefits of "Communication

through Writing, Typing, and Verbal Expression” and “Social Inclusion and Connection.” While the former emphasized expressive abilities across modalities, the latter extended communication into relational contexts, highlighting how technologies facilitated connection, belonging, and participation in social life. A similar relationship appeared within the challenges: “Interface and Interaction Complexity” and “Contextual Misalignment in Technology Use” were often linked, where difficulties with layout, responsiveness, or navigation reflected not only technical limitations but also a disconnect between how systems were designed and how AwDS expected or preferred to engage with them. Themes also bridged benefits and challenges, such as “Skill Development and Learning” and “Limited Generalization and Uneven Impact.” Many studies documented measurable learning gains, but these were often paired with concerns about variability in outcomes or a lack of sustained progress across users. These patterns underscore the heterogeneity of technology experiences within this population, shaped not only by user preferences and contexts, but also by the limited sample sizes and short durations of many studies. While we initially expected independence to be the dominant framing, as is common in assistive technology research, we were pleasantly surprised to find that many studies paired autonomy with connection. Independence remained a prominent theme, especially within non-educational tools designed to support daily routines, navigation, or digital participation. Yet across much of the literature, autonomy was not framed in isolation, but in relation to social connection. Technologies often supported users in managing tasks on their own terms while remaining embedded in social networks, whether by enabling communication, coordinating with caregivers, or sharing experiences with friends and family. This recurring pairing of independence and connectedness reflects a more relational view of autonomy, one that resonates strongly with the lived experiences of many AwDS. Rather than aiming solely for self-sufficiency, the technologies reviewed here often sought to foster meaningful participation and belonging, reinforcing the importance of designing for interdependence as well as individual ability.

Design Recommendations. Five design recommendations emerged from the reviewed studies, emphasizing the importance of creating technologies that adapt to the needs, preferences, and contexts of AwDS. First, reducing interaction complexity through simplified visuals, intuitive layouts, and minimal on-screen options supported navigation and task completion across learning, communication, and navigation tools [79, 85]. Second, systems that offered multisensory feedback, flexible pacing, and user-controlled interaction (e.g., pre-recorded voice prompts, animated cues, or pause-and-repeat options) promoted sustained engagement and independent learning [13, 70]. Third, personalization features and adaptive content supported meaningful use and long-term growth, enabling users or caregivers to adjust routines, difficulty levels, or motivational rewards over time [29, 88]. Fourth, secure access pathways, such as user-friendly password formats, biometric options, or caregiver-assisted setup, helped reduce barriers to system entry while maintaining safety and privacy [92, 99]. However, it is important to note that caregivers may require their own training and demo session, and AwDS may not want to include them in the use of their systems. Finally, technologies were most successful when designed to function within everyday environments, supporting use across home, work, and public spaces through GPS tracking, visual cues, and offline functionality [79, 98]. Together, these strategies highlight a shift toward inclusive systems that meet users where they are, embedding support into familiar routines.

The reviewed studies suggest a gradual reorientation from technologies that are education-focused to those that are contextually practical, shifting focus from skill acquisition toward everyday participation, digital autonomy, and real-life routines (as seen in literature with children with DS [59, 67, 84]). Across this work, social connection emerged as a consistent thread with many tools supporting independence in relation to social connectedness. Many studies were not aimed at

promoting social connection; however, they discussed how these tools still facilitated connection, even when focused on independence itself. Social connectedness is not a new concept in accessibility [14, 115]. Still, its emphasis here reflects a shift from prior technology literature, particularly when compared to studies involving other disability communities [3, 81, 105, 130, 143, 153]. Individuals with DS are often stereotyped as having strong social motivation and skills, in contrast to other groups. Yet, AwDS may still experience infrequent peer interaction and limited social opportunities in adulthood [46]. Future research should embrace this distinction by shifting focus from pure independence to embedding connection into everyday technologies. That is, designing tools that foster social participation and meaningful interaction, even when their primary purpose is not explicitly social. This orientation reflects a more relational model of autonomy, where tools support both self-directed action and ongoing connection in the daily lives of AwDS. To ensure these technologies are genuinely meaningful, AwDS must be included as active contributors in the design process. While stakeholders can offer important context, centering the voices of AwDS affirms their agency and results in tools that are not only functional but also personally valuable. This review lays the groundwork for future research, emphasizing the need for user-centered, adaptable designs that reflect the full range of abilities, goals, and preferences within the AwDS community.

7 Future Work

Future research should continue to center AwDS in designing, evaluating, and developing technologies. This means moving beyond broad accessibility claims to create solutions developed in collaboration with the community that reflect not only what technologies should do, but also why they matter. Future studies should explore diverse motivations for technology use, such as personalization, digital autonomy, engagement, and identity expression, and prioritize those that align with users' everyday goals. Researchers should also examine the contexts in which support, whether through technology or people, is welcomed or intrusive. For instance, an AwDS user may rely on assistive tools to read text messages, but still wants privacy when receiving messages from a significant other while around family.

Geographic representation also emerged as a notable limitation. Most reviewed studies were conducted in North America and Europe, with no included papers from regions such as South America, Africa, or Asia (excluding one publication from Iran). This lack of cultural diversity limits the generalizability of findings. It raises questions about how social norms, infrastructure, and local values influence access to and the design of technology for AwDS. Future work should prioritize more inclusive sampling, localized research approaches, and culturally sensitive design practices that reflect global diversity in lived experience and technological contexts. More work could also be done on the complex contextual relationship between AwDS and their support systems, as well as on how technology can be used independently or as a support mechanism between groups.

One critical direction is enhancing personalization. Many current technologies offer limited ability to adjust difficulty, pacing, or interface elements to match the diverse cognitive and motor profiles of AwDS. Future systems should offer layered levels of support, enabling users to toggle between guided and independent modes or adjust complexity over time. Tools such as large language models and generative AI present new opportunities for real-time adaptation. For example, personalized avatars, simplified explanations, or dynamically generated task instructions could help tailor experiences to individual users while preserving autonomy and engagement. AI-driven feedback loops could further adjust systems based on user progress, offering gentle guidance or encouragement aligned with user goals.

Employment-related technologies also warrant expanded exploration. While studies have explored office-based tools, future work should consider technologies that support employment in other industries, such as retail, hospitality, the arts, or service roles. Wearables, robots, and

AR-based systems may be particularly useful for on-the-job training, task sequencing, reminders, collaboration, or spatial navigation in dynamic environments. Research should explore ways to support AwDS in remote or hybrid work arrangements, particularly in light of the increasing decentralization of the workforce. Additionally, financial systems and their data visualizations should be explored to understand how designers can make them more accessible for AwDS.

Digital safety remains a pressing and underexplored area. AwDS face unique risks related to password management, privacy, and exposure to inappropriate content. Future research should investigate how to support safe online behavior through accessible education materials, such as visual tutorials or interactive simulations. Authentication systems and privacy prompts should also be reviewed and redesigned without compromising security (e.g., biometric tools, simplified two-factor authentication, or secure caregiver-assisted options). Real-time support systems could further help AwDS navigate confusing or risky digital scenarios and confidently recover from mistakes.

In addition to screen-based tools, future work should expand to tangible and embodied technologies. TUIs, which involve interacting with physical objects, may offer intuitive alternatives to abstract interfaces. TUIs could support daily routines, therapy, or creative expression by embedding interaction into familiar, sensory-rich contexts (e.g., smart textiles or smart tabletops). These tools may be particularly beneficial for users who benefit from hands-on learning or struggle with multistep digital interfaces. Alternatives to cognitively demanding interface structures, such as multilayered menus, should be explored. Simplified navigation models, including gesture-based navigation, voice commands, customizable home screens, or context-aware shortcuts, could support faster and more confident engagement with complex systems. TUIs and physical interfaces may offer another route, especially when abstract navigation proves overwhelming. These design approaches should aim to maintain the functionality of complex systems while providing simpler ways for users to interact with them, thereby reducing errors and enhancing usability.

Finally, speech and gesture recognition technologies also need refinement. Current systems often fail to accommodate the speech patterns of AwDS, leading to user frustration and abandonment. Research is necessary to improve recognition models by utilizing inclusive training datasets and adaptive calibration processes that account for individual variability. Just as importantly, these systems should offer meaningful feedback and alternatives when recognition fails so that users feel supported, not penalized. Taken together, these directions underscore a broader call for research that is participatory, context-aware, and ability-focused. Technologies designed with AwDS must reflect the realities of everyday life, adapt to evolving needs, and affirm the autonomy and perspectives of users at every step.

8 Limitations

The scope of this literature review is limited by the relatively small number of publications typically expected for a systematic review. Only 20 publications specifically studied technology with AwDS, and as a result, the findings do not fully represent the diversity of abilities, cultures, and preferences within this population. Some relevant studies may have been missed if they were hosted on databases we did not search or if our broad search terms still failed to capture all applicable literature. Additionally, our inclusion criteria relied on authors explicitly stating that their participants were AwDS. It is possible that other studies included AwDS participants but did not specify “Down syndrome” or age demographics clearly enough for inclusion. As a small research team conducted this review, the processes of screening, inclusion, and interpretation may have been influenced by subjective judgment despite the use of structured methods. Due to this being a scoping review, we did not perform a risk of bias on the included publications, as recommended by PRISM-ScR guidelines, which could introduce bias in the results. We also did not collect data on IRB compliance

within the papers selected for review. Finally, we only included English-language, peer-reviewed publications, which may exclude valuable insights from international, community-led, or grey literature sources.

9 Conclusion

This scoping literature review synthesized findings from 20 peer-reviewed publications that examined technologies designed and used by AwDS. The goal of this review was to gain a deeper understanding of how technology is being developed and studied with this population. We explored (1) how researchers have involved AwDS in the design and evaluation of technologies, including inclusive methods and accessibility strategies; (2) the technologies that have been explored and the motivations and perceived advantages driving their use; (3) how digital technologies have been applied to enhance abilities in various life domains; and (4) the reported benefits and challenges experienced by AwDS when interacting with these technologies. Our analysis identified five key benefits: confidence, enjoyment, and engagement; independence and digital inclusion; skill development and learning; communication through writing, typing, and verbal expression; and social inclusion and connection. We also identified four common challenges: interface and interaction complexity; contextual misalignment in technology use; limited generalization and uneven impact; and environmental and sensory constraints. We additionally outlined five design recommendations that emphasize the importance of: (1) reducing interaction complexity through clear visuals and simplified choices; (2) supporting engagement through responsive feedback and user-controlled pacing; (3) embedding design into everyday routines and environmental contexts; (4) designing for individualization, adaptability, and growth over time; and (5) Provide flexible, user-friendly pathways to secure system access. These insights underscore the need for inclusive, adaptive, and context-aware technologies that align with the preferences, abilities, and goals of AwDS. By centering their lived experiences and capabilities, future research and design efforts can advance toward more meaningful, empowering, and equitable use of technology.

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Appendices

A Metadata Table

Table A1. Summary of Meta-Data for All 20 Included Studies

PID	Citation	Publication Year	Country	Continent	Field Code
P01	[85]	2012	USA	Americas	Communication Sciences and Disorders; Computer and Information Sciences
P02	[4]	2019	Iran	Asia	Special Education and Psychoeducation
P03	[92]	2018	USA	Americas	Computer and Information Sciences; Healthcare and Health Sciences
P04	[13]	2014	Italy	Europe	Healthcare and Health Sciences; Computer and Information Science
P05	[20]	2015	Hungary	Europe	Engineering
P06	[98]	2018	Canada	Americas	Special Education and Psychoeducation; Engineering; Community Group
P07	[70]	2023	USA	Americas	Communication Sciences and Disorders; Special Education and Psychoeducation
P08	[104]	2012	USA	Americas	Special Education and Psychoeducation
P09	[150]	2021	USA	Americas	Communication Sciences and Disorders
P10	[68]	2022	USA	Americas	Communication Sciences and Disorders
P11	[99]	2013	USA	Americas	Computer and Information Sciences; Communication Sciences and Disorders
P12	[29]	2016	Canada	Americas	Computer and Information Sciences; Engineering
P13	[39]	2015	Italy	Europe	Engineering; Community Group
P14	[54]	2002	Australia	Oceania	Special Education and Psychoeducation
P15	[148]	2023	Spain	Europe	Computer and Information Sciences
P16	[77]	2022	Norway	Europe	Healthcare and Health Sciences
P17	[79]	2021	Scotland	Europe	Computer and Information Sciences
P18	[91]	2011	USA	Americas	Computer and Information Sciences; Communication Sciences and Disorders
P19	[88]	2011	Canada	Americas	Special Education and Psychoeducation
P20	[89]	2019	France	Europe	Healthcare and Health Sciences

Each row contains the unique paper ID (PID), citation reference, year of publication, country and continent of origin, and associated disciplinary field(s). Field codes reflect the primary domains represented in each study, including communication sciences, computer science, health sciences, education, and engineering.

B Demographics Table

Table B1. Participant Demographic Data for Each of the 20 Included Studies

PID	Citation	Low Age	High Age	Average Age	Number of Fe-male	Number of Male	Sample Size
P01	[85]	19	29	22.9	6	4	12
P02	[4]	25	36	30	4	6	10
P03	[92]	18	30	21.5	5	5	10
P04	[13]	-	-	-	-	-	2
P05	[20]	-	-	-	-	-	0
P06	[98]	-	-	-	-	-	15
P07	[70]	18	27	-	0	3	3
P08	[104]	29	35	-	0	4	4
P09	[150]	20	24	21.6	2	1	3
P10	[68]	21	39	-	4	2	6
P11	[99]	18	39	28	8	2	10
P12	[29]	28	32	30	1	1	2
P13	[39]	30	30	30	1	0	1
P14	[54]	19	19	19	1	0	1
P15	[148]	23	36	-	2	5	7
P16	[77]	22	22	22	5	3	8
P17	[79]	18	28	22.1	4	3	7
P18	[91]	22	38	-	7	3	10
P19	[88]	19	19	19	0	1	1
P20	[89]	21	44	26.2	2	10	12

The table reports the minimum and maximum ages (Low Age and High Age), average age when available, number of female and male participants, and total sample size per study. Missing data are indicated with dashes. This table highlights the diversity in participant ages and gender representation across studies.

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C Methods and Technology Table

Table C1. Overview of Study Methodologies, Inclusive Strategies, Technologies Explored, and Reported Outcomes Across the 20 Included Studies

PID	Citation	Study Methods	Inclusive Approaches	Data Collection	Technologies Explored	Technology Advantages	Application of Technology	Enhanced Ability through Technology
P01	[85]	Usability	Stakeholder or expert involvement; Study customization and personalization; Study instruction support; Visual study aids; Preparation and training with the technology	Mixed	Mobile technologies	Support individuals in accessing opportunities	Support; Digital inclusion	Workplace assistance
P02	[4]	Experimental	Stakeholder or expert involvement; Ensuring comfortable environment; Visual study aids	Quantitative	Social robotics	Support education or therapy; Enhance engagement or motivation	Teach	Cognitive, memory, and learning assistance; Communication and literacy
P03	[92]	Participatory design	Stakeholder or expert involvement; Study customization and personalization; Ensuring comfortable environment; Visual study aids	Qualitative	Mobile technologies	Support individuals in accessing opportunities; Enable faster personalization or customization; Enhance engagement or motivation	Support; Teach	Health and nutrition
P04	[13]	Usability	Stakeholder or expert involvement	Qualitative	Mobile technologies	Support education or therapy; Enable faster personalization or customization; Enhance engagement or motivation	Practice	Cognitive, memory, and learning assistance
P05	[20]	Development	-	None	Computers and computer-based web technologies	Support education or therapy; Enhance engagement or motivation	Teach	Cognitive, memory, and learning assistance
P06	[98]	Exploratory; Participatory design	-	Qualitative	Broader digital inclusion	Support individuals in accessing opportunities	Digital inclusion	Independent living
P07	[70]	Experimental	-	Mixed	Computers and computer-based web technologies; AAC	Support education or therapy; Enable faster personalization or customization	Teach	Communication and literacy

(Continued)

Table C1. Continued

PID	Citation	Study Methods	Inclusive Approaches	Data Collection	Technologies Explored	Technology Advantages	Motivations or Application of Technology	Enhanced Ability through Technology	Enhanced Technology
P08	[104]	Experimental	Study instruction support	Qualitative	Computers and computer-based web technologies	Support education or therapy	Practice	Independent living	
P09	[150]	Experimental	Stakeholder or expert involvement; Study customization and personalization; Study instruction support	Qualitative	Computers and computer-based web technologies; AAC	Support individuals in accessing opportunities	Support	Communication and literacy	
P10	[68]	Exploratory	Stakeholder or expert involvement; Study instruction support; Ensuring comfortable environment; Visual study aids	Qualitative	Computers and computer-based web technologies; AAC	Support individuals in accessing opportunities	Support; Digital inclusion	Communication and literacy	
P11	[99]	Usability	-	Qualitative	Computers and computer-based web technologies	Support individuals in accessing opportunities	Digital inclusion	Independent living	
P12	[29]	Usability	Study instruction support; Visual study aids	Mixed	Computers and computer-based web technologies; AAC; Mobile technologies	Support education or therapy; Enable faster personalization or customization; Enhance engagement or motivation	Practice	Cognitive, memory, and learning assistance	
P13	[39]	Observational and ethnographic	Study instruction support	Mixed	Smart home systems	Support individuals in accessing opportunities; Enable faster personalization or customization	Support; Teach	Independent living	
P14	[54]	Experimental	Stakeholder or expert involvement; Study customization and personalization	Mixed	Computers and computer-based web technologies	Support education or therapy; Enhance engagement or motivation	Teach	Communication and literacy	

(Continued)

Table C1. Continued

PID	Citation	Study Methods	Inclusive Approaches	Data Collection	Technologies Explored	Technology Advantages	Motivations or Application of Technology	Ability through Technology	Enhanced Technology
P15	[148]	Exploratory	Stakeholder or expert involvement	Mixed	Mobile technologies	Support education or therapy; Enhance engagement or motivation	Teach	Cognitive, memory, and learning assistance; Health and nutrition	
P16	[77]	Exploratory	Stakeholder or expert involvement; Study customization and personalization; Ensuring comfortable environment	Qualitative	Broader digital inclusion	Support individuals in accessing opportunities	Digital inclusion	Independent living	
P17	[79]	Participatory design	Ensuring comfortable environment	Qualitative	Mobile technologies	Support individuals in accessing opportunities; Enable faster personalization or customization	Teach	Independent navigation	
P18	[91]	Observational and ethnographic	-	Qualitative	Computers and computer-based web technologies	Support individuals in accessing opportunities	Teach	Workplace assistance	
P19	[88]	Usability	Stakeholder or expert involvement; Study customization and personalization	Qualitative	Mobile technologies	Support individuals in accessing opportunities; Enable faster personalization or customization	Teach	Independent navigation	
P20	[89]	Experimental	-	Qualitative	Computers and computer-based web technologies	Assess participants	Practice	Independent navigation	

The table presents each paper's methodological approach, inclusive research practices, data collection type, and the technologies examined. It also highlights the motivations for using technology, how the technologies were applied, and the specific abilities enhanced through their use.